Do Prices Reveal What Insiders Know? A Reexamination of Plott and Sunder (1982)*

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Abstract

I reexamine a seminal laboratory study reporting that, with repetition, prices quickly reveal what insiders know. In its design, prices can distinguish rational expectations from prior information only in downward periods, when insiders' information lowers the asset's value below what uninformed traders would pay; when it raises the value, both theories predict the same price. Of the 17 downward periods behind the original conclusion, only those in one of five markets show prices moving from their no-information level toward the fully revealing price. In two markets, prices are already near that price when no one is informed; in the other two, they stay above it. In upward periods, the first trade covers about 30 percent of the distance to the informed value, and prices close most of the gap during trading. As a pretest for a human experiment, I run LLM-agent markets that make the two directions symmetric; prices there cover only about 40 percent of the distance on both sides.

Keywords: insider trading; rational expectations; experimental asset markets; price formation; LLM agents
JEL codes: C92; D82; D84; G14

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Email: siruizou2005@gmail.com. Data and code are available at https://github.com/siruizou2005/Plott-Sunder-1982.

1 Introduction

Whether prices can pass information held by a few traders on to everyone else is central to debates about market efficiency and the regulation of markets. The strong form of the efficient markets hypothesis holds that prices fully reflect even information to which only some investors have monopolistic access (Fama, 1970, p. 409). Theory gives two answers to whether they do. Prices can reveal the informed traders' information to all (Grossman, 1976), or they may incorporate an insider's information only gradually (Kyle, 1985). Field data rarely settle the matter, because researchers seldom observe who is informed; one of the few direct sources is insider-trading investigations by the Securities and Exchange Commission, which document trades made on nonpublic information (Kacperczyk and Pagnotta, 2019).

Laboratory markets supply what field data lack. The experimenter assigns each trader's value, so the equilibrium is known and observed prices can be compared with it (Chamberlin, 1948; Smith, 1962); the experimenter also decides who knows what. Plott and Sunder (1982) used this control to study insider information. In their markets, traders buy and sell certificates of a one-period asset whose dividend depends on a randomly drawn state and differs across types of traders. Before trading, some traders, the insiders, learn the state; the others do not, and they do not know who is informed. Trading takes place through publicly announced bids and offers. Each trader starts every period with two certificates and ample cash, and short sales are not allowed. Two theories give competing predictions for prices. Under rational expectations (RE), uninformed traders infer the state from the market, so the price reaches the value implied by the state. Under prior information (PI), they do not, and they continue to value the asset at its expected dividend under the prior.

Plott and Sunder's abstract reports that "with replication of market conditions the predictions of a fully revealing rational-expectations model are relatively accurate" and that "prices adjusted immediately to near rational-expectations prices" (p. 663). Later work reads the study as evidence that markets carry insiders' information into prices. Gillen et al. (2017, p. 1079) write that it "first demonstrated the ability of continuous double auction markets to transfer information from 'insiders'

who have information about the state to noninsiders who do not." Crockett et al. (2019, p. 628) describe early experiments, this one among them, as producing prices that "effectively aggregate private dividend information and converge towards rational expectations values." Choo et al. (2019, p. 627) summarize the finding as "asset prices converge to fully reveal the true state."

I reexamine the price evidence behind this conclusion. The conclusion rests on comparing observed prices with the RE and PI predictions, and the two predictions do not differ in every period. When insiders learn that the asset is worth more than uninformed traders believe, the two theories predict the same price: insiders, who have ample cash, are the highest bidders whether or not anyone else learns. When insiders learn that it is worth less, they can sell only the certificates they hold, so the predictions diverge: under PI, uninformed buyers keep bidding their prior value; under RE, they infer the news from the market, and the price falls to the informed value. I call these downward periods and the others upward periods. In the original design, the two predictions coincide in every upward period, whatever the dividends and priors, so only downward periods can test the two theories. Of the 17 downward periods behind the original price conclusion, spread over five markets, prices move from their no-information level toward RE in only one market. In two markets, prices were already near RE when no one was informed; in the other two, they stayed above RE. In upward periods, where information has to move prices, prices adjust gradually within the trading period.

Two pieces of evidence support this reading, both drawn from the trades listed in the appendix of the working-paper version of the study (Plott and Sunder, 1980). In two of the markets, downward-period prices are almost the same as prices when no one is informed (in one of them, 163 francs in no-information periods and 164 in downward periods), and both lie 35 to 47 francs below the prior value. A comparison of downward-period prices with the two theories therefore cannot separate prices that incorporate insiders' information from prices that were already there. In the one market where prices do move, the no-information price is near the prior value, and downward-period prices fall to near the RE price. In the upward periods of the three markets with similar designs, the first trade covers only about 30 percent of

the distance from the prior value to the informed value, and most of the adjustment takes place during the trading period.

The paper relates to three lines of work. Studies after Plott and Sunder (1982) varied the number of insiders, often without telling traders how many there were, to ask whether and how quickly prices reveal what insiders know (Camerer and Weigelt, 1991; Bossaerts et al., 2014; Stöckl, 2014). Plott and Sunder (1988) turned from dissemination to aggregation, in which information is dispersed across traders and no single trader knows the state; later work compared concentrated with dispersed information (Corgnet et al., 2020) and measured how much private information prices incorporate (Page and Siemroth, 2021). Corgnet et al. (2023) reexamined the 1988 results with a high-powered replication, which largely failed; dispersed information reliably entered prices only in new markets that combine identical dividends across traders with a complete set of contingent claims. Finally, replicating Plott–Sunder-style experiments with artificial agents has a precedent (Chan et al., 1999). Large language models (LLMs) have been used to simulate experimental subjects and to predict experimental results (Aher et al., 2023; Horton et al., 2023; Manning et al., 2024; Ashokkumar et al., 2026; Manning and Horton, 2025), and they have been placed in markets and games (Shah et al., 2025; del Rio-Chanona et al., 2025; Wang et al., 2025; Qian et al., 2026).

Why do upward periods adjust slowly? In the original design, the direction of adjustment comes bundled with three other conditions: how far prices must travel, whether insiders agree on which way to trade, and whether uninformed traders compete at the target price. The existing data cannot separate them. As a pretest, I run markets in which every trader is an LLM agent and the two directions are symmetric in these conditions. In these markets, prices cover on average only about 40 percent of the distance in both directions. The pretest points to a human experiment that is symmetric between buying and selling and in which both directions can distinguish RE from PI, which I leave to future work.

The paper adds to each of these lines. For experiments with insiders, it shows that the classic price test can distinguish RE from PI only in downward periods, a property of the original design that holds for any dividends and priors. For the reexamination

of classic evidence, it complements Corgnet et al. (2023), who replicate the 1988 study: I run no new human sessions and instead use the original experiment's own records to add two comparisons the original does not report, downward-period prices against no-information prices and the price paths within upward periods. For the LLM literature, it uses LLM markets to pretest the design of a human experiment.

Section 2 describes the original experiment and defines three price levels. Section 3 derives when the RE and PI predictions differ and lists the other conditions that change with the direction of adjustment. Section 4 compares downward-period prices with no-information prices. Section 5 traces prices within upward and downward periods. Section 6 reports the LLM pretest, and Section 7 concludes.

2 The Original Experiment and Three Price Levels

2.1 Market environment

Plott and Sunder (1982) ran five markets, each with 9 or 12 traders divided into three dividend types, I, II, and III (Table 1). In each period a state θ was drawn at random, and it fixed the dividend $d_t(\theta)$ that a certificate paid to a trader of type t. The prior probabilities $\pi(\theta)$ of the states were public. In an insider period, some traders of each type received a clue card naming the state before trading, two of four in markets 2–5 and one of three in market 1; the others received blank cards (Plott and Sunder, 1982, p. 667).[1] No trader knew how many traders were informed or which ones, and no trader knew the other types' dividends (p. 670). Appendix Table A2 lists every dividend.

Trading was an oral double auction in which each accepted bid or offer transferred one certificate (Plott and Sunder, 1982, pp. 666, 695). At the start of every period each trader held two certificates and 10,000 francs, and holdings were reset at the end

[1] In market 1 the insiders saw a ten-draw sample that is informative about the state, not the state itself. I classify its periods with the values conditional on the sample; Appendix A.4 gives the details.

Table 1: The five original markets

Market	Traders	Insiders per type	States	Prior $\pi(\theta)$	Periods		
					No-information	Insider	All-informed
1	9	1 of 3	2	X: 1/3, Y: 2/3	1–4	5–8	9–11
2	12	2 of 4	2	X: 1/3, Y: 2/3	1–4	7–11	5–6
3	12	2 of 4	2	X: 0.40, Y: 0.60	1–2	3–10	11–12
4	12	2 of 4	2	X: 0.40, Y: 0.60	1–4, 14	5–13	–
5	12	2 of 4	3	X: 0.35, Y: 0.25, Z: 0.40	1–3	4–13	–

Notes: Insiders per type is the number of traders of each dividend type who learn the state in an insider period, out of the number of traders of that type. Source: Plott and Sunder (1982, Tables 1–2).

of the period. Short sales were prohibited, so traders could sell only the certificates they held, and working capital was "sufficiently large never to serve as a binding constraint on purchases of securities" (p. 666). At the end of a period each certificate paid its holder's dividend in the realized state, and a fixed cost of 10,000 francs was deducted (pp. 666, 670).

Of the 61 periods, 18 informed no one, 7 informed everyone, and 36 were insider periods.

2.2 Three price levels

Market 4 illustrates the three price levels (Table 2). An uninformed trader values a certificate at its expected dividend under the prior. The highest such value belongs to type I, at 210 francs, so the prior value in market 4 is $\mu = 210$. Once the state is known, the highest dividend is 375 francs in state X and 175 in state Y; these are the informed values. Panel B also lists the PI prediction for each state. Section 3.1 uses supply and demand to explain why the two predictions differ in state Y and coincide in state X.

In general, the prior value is the highest expected dividend under the prior,

$$\mu = \max_t E[d_t], \qquad E[d_t] = \sum_\theta \pi(\theta)\, d_t(\theta), \tag{1}$$

Table 2: Market 4: dividends, prior values, and price levels

Panel A. Dividends and prior expected values

Type	State X ($\pi = 0.4$)	State Y ($\pi = 0.6$)	Prior expected value
I	375	100	210
II	275	150	200
III	100	175	145

Panel B. Price levels

State	$P_{\mathrm{RE}}(\theta)$	$P_{\mathrm{PI}}(\theta)$
X	375	375
Y	175	210

Notes: Francs per certificate. The prior value is $\mu = \max_t E[d_t] = 210$. Source: Plott and Sunder (1982, Tables 1–3); author's calculations.

the most any uninformed trader would pay. The informed value is the highest dividend once the state is known,

$$P_{\mathrm{RE}}(\theta) = \max_t d_t(\theta), \tag{2}$$

which is also the RE price prediction. The PI prediction, $P_{\mathrm{PI}}(\theta)$, is the highest value in a market in which insiders value a certificate at its realized dividend and uninformed traders value it at its expected dividend under the prior; Section 3.2 derives its form. Both predictions rest on the original assumptions that traders are risk neutral and that the price equals the value of the highest bidder (Plott and Sunder, 1982, Section III). I follow the original in writing θ for the state and d for dividends, index dividends by type because traders of a type receive the same dividend, and write the prior as $\pi(\theta)$ because p denotes prices. Upper-case P and μ are predictions; lower-case p is an observed price.

3 The Asymmetry in the Original Design

3.1 Who sets the price under the original rules

The rules in Section 2.1 shape how risk-neutral traders behave. A trader facing a price below the trader's own value buys as many certificates as possible, and because capital is ample, the quantity is unlimited; facing a price above that value, the trader sells, but no more than the two certificates in hand. Market demand is therefore horizontal at the highest value in the market, and the price equals the value of the type that values a certificate most, several of whose members compete with one another. In Plott and Sunder's words, "the demand price will equal the expected dividend value of the agent type with the maximum expected dividend" (p. 672). This is the pricing convention behind equation (2) and the PI prediction.

Consider the two states of market 4 (Table 2). In state Y, the insiders' values are 100, 150, and 175 francs, all below 210, so every insider is a seller. The highest bidders are the uninformed type I traders, who bid their prior value of 210. If the uninformed do not learn the state, the price stays at 210, the PI prediction. If they learn it, the type I value falls to 100, and the price falls to the highest informed value, 175, the RE prediction. Whether the uninformed learn the state therefore changes the price. In state X, the highest bidders are the two informed type I traders, who value a certificate at 375 and compete with each other, so the price is 375. If the uninformed do not learn the state, their highest value is 210 and they sell, and the price is still 375. If they learn it, the uninformed type I traders also bid up to 375, and the price is again 375. Whether the uninformed learn the state does not change the price.

State Y is a *downward period*: an insider period in which $P_{\mathrm{RE}}(\theta) < \mu$. State X is an *upward period*: an insider period in which $P_{\mathrm{RE}}(\theta) > \mu$. Because no insider values a certificate above $P_{\mathrm{RE}}(\theta)$, a downward period is one in which every insider would sell at the prior value.

3.2 Proposition 1 and period-by-period verification

The argument for market 4 holds in general under four conditions:

(A1) Insiders know the realized state. The clue names the state, as it does in markets 2–5.

(A2) Under PI, uninformed traders value a certificate at its prior expected dividend; under RE, the price reveals the state. These are the two hypotheses being compared.

(A3) In an insider period, every type has both insiders and uninformed traders. This is the key condition, and the design builds it in: "In most periods of all markets only one-half (two out of four) of the agents from each dividend (preference) type received information about the realized state of nature" (p. 667).

(A4) Traders are risk neutral, the price equals the highest bidder's value, and buyers' capital can absorb the whole supply. This is the original pricing convention; one trader's 10,000 francs buy 25 certificates even at 400 francs, the highest informed value in the design, against a supply of at most 24 certificates.

Proposition 1. *Under (A1)–(A4), the RE and PI price predictions are equal in every upward period, whatever the dividends and priors; they differ only in downward periods.*

Proof. Under RE, the price reveals the state (A2), so every trader values a certificate at the realized dividend of the trader's type, and by (A4) the price is the highest of these dividends, $P_{\mathrm{RE}}(\theta)$. Under PI, insiders value a certificate at its realized dividend (A1) and uninformed traders at their prior expected dividend (A2), so no value in the market exceeds $\max\{P_{\mathrm{RE}}(\theta), \mu\}$. By (A3), both terms are present: the type with the highest realized dividend has an insider, and the type with the highest prior value has an uninformed member. By (A4), the price equals this maximum:

$$P_{\mathrm{PI}}(\theta) = \max\{P_{\mathrm{RE}}(\theta), \mu\}. \tag{3}$$

Hence $P_{\mathrm{PI}}(\theta) = P_{\mathrm{RE}}(\theta)$ when $P_{\mathrm{RE}}(\theta) > \mu$, as in state X of market 4, and $P_{\mathrm{PI}}(\theta) = \mu > P_{\mathrm{RE}}(\theta)$ when $P_{\mathrm{RE}}(\theta) < \mu$, as in state Y, so the two predictions differ if and only if $P_{\mathrm{RE}}(\theta) < \mu$. □

Condition (A3) is what makes upward periods uninformative. If no trader of the type with the highest dividend in an upward state were an insider, that upward period could distinguish the two theories, because prices could reach P_{RE} only if uninformed traders of that type learned the state from prices. In state X of market 4, for example, if no type I trader were informed, the PI prediction would be $\max\{275, 210\} = 275$ against an RE prediction of 375.

Plott and Sunder note that the two predictions differ when the uninformed traders' prior value lies above the RE price (p. 673). Proposition 1 shows that, in their design, this is the only case: under (A3), no upward period can separate the two theories, whatever the dividends and priors.

The proposition is a statement about parameters, so I check it period by period rather than test it. I recompute μ, P_{RE}, and P_{PI} for every period from the priors, dividends, and information schedules in Tables 1 and 2 of Plott and Sunder (1982) (Appendix Table A3). Of the 61 periods, 36 are insider periods, 17 downward and 19 upward (Table 3). The two predictions differ in all 17 downward periods and coincide in all 19 upward periods (Figure 1). Plott and Sunder list the same 36 periods (p. 683) and base their price test on the 17 in which the two predictions differ (p. 679, n. 6). Market 1 is classified with values conditional on the insiders' sample (Corollary 1 in Appendix A.4). The original footnote 6 lists period 7 instead of period 6 among market 3's price-test periods; the recomputation and the original Table 5 both point to period 6, and the count of 17 is unchanged (Appendix B.3).

Of the original tests, the price test and the efficiency test are confined to downward periods; the original efficiency table shows 15 of the 17. The allocation test covers all 36 insider periods, and the profit comparison covers both kinds of period (pp. 683–685; Appendix A.5).

Table 3: The 61 periods by information condition and period type

				Insider periods	
Market	Periods	No-information	All-informed	Downward	Upward
1	11	4	3	2	2
2	11	4	2	2	3
3	12	2	2	5	3
4	14	5	0	5	4
5	13	3	0	3	7
Total	61	18	7	17	19

Notes: Insider periods inform a strict subset of traders. A downward period has $P_{\mathrm{RE}} < \mu$ and an upward period $P_{\mathrm{RE}} > \mu$, with μ and P_{RE} computed from the priors and dividends. By Proposition 1, the RE and PI predictions differ in every downward period and in no upward period. Market 1 uses values conditional on the insiders' ten-draw sample (Appendix Table A4). Source: information schedules and parameters in Plott and Sunder (1982); author's calculations.

3.3 Other differences between downward and upward periods

The analysis of Section 3.1 shows three further ways in which the two kinds of period differ in markets 3–5 (Table 4). First, prices must travel farther in upward periods. In market 3, the informed value lies 180 francs above the prior value in upward periods and 45 francs below it in downward periods; in market 4 the distances are 165 and 35 francs. Market 5 is an exception, since some of its upward periods require only 32.5 francs.

Second, insiders agree on the direction of trade only in downward periods. At the prior value, all six insiders want to sell in a downward period. In the upward periods of markets 3 and 4, four want to buy and two want to sell; in those of market 5, two want to buy and four want to sell. Insiders who keep selling in an upward period add supply and may slow the rise in prices.

Third, in downward periods some uninformed traders can trade at the target price without learning anything: their prior value lies below P_{RE}, so they are willing to sell at it. There are two such traders in markets 3 and 4 and four in market 5. No uninformed trader is in this position in an upward period, since every prior value

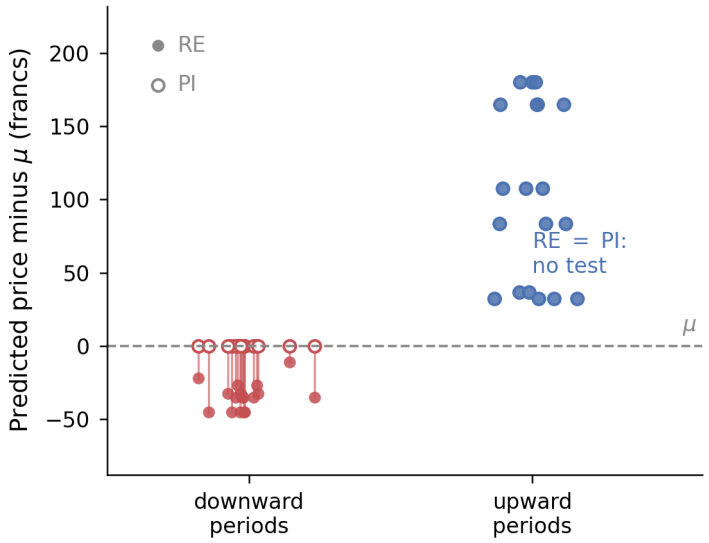


Figure 1: The RE and PI predictions differ only in downward periods. Each marker is one of the 36 insider periods, placed at its predicted price minus the prior value μ, in francs. Filled markers are RE predictions and open markers PI predictions; a segment joins them where they differ. In the 19 upward periods the predictions coincide and print as one marker. Markers are spread horizontally within each group for legibility. Market 1 uses values conditional on the insiders' sample. Source: market parameters in Plott and Sunder (1982); author's calculations.

lies below P_{RE}.

Table 4: Differences between upward and downward periods in markets 3–5

	Distance $P_{\mathrm{RE}} - \mu$		Insiders buying minus selling at μ		Uninformed trading at P_{RE} at prior value	
Market	Upward	Downward	Upward	Downward	Upward	Downward
3	+180	−45	+2	−6	0	2
4	+165	−35	+2	−6	0	2
5	+32.5 or +107.5	−32.5	−2	−6	0	4

Notes: Distance is $P_{\mathrm{RE}} - \mu$ in francs; market 5 has two upward states. Insiders buying minus selling counts, at a price of μ, insiders whose value exceeds μ minus insiders whose value is below it (two insiders per type). The last pair of columns counts uninformed traders whose prior value lets them trade at P_{RE} in the direction of adjustment without learning the state. Source: Plott and Sunder (1982, Tables 1–2); author's calculations.

A fourth difference comes from the trading rules. Sellers can sell at most the two

certificates they hold, while buyers have ample capital, so insiders on the two sides can trade different quantities and earn different rents. Like the equality of the RE and PI predictions in upward periods, this difference follows from the rules, and no choice of dividends or priors removes it.

In the original design, all these differences move together with the direction of adjustment, and the human data cannot separate them. The comparisons of downward and upward periods in Sections 4 and 5 therefore describe differences without attributing them to direction itself. Separating them requires a symmetric design, which Section 6 pretests.

4 Price Evidence from Downward Periods

4.1 The no-information benchmark

In a no-information period nobody is informed, so RE and PI both predict a price equal to the prior value μ. I define a market's no-information price $\bar{p}_0$ as the average, over its no-information periods, of each period's mean transaction price. The no-information periods are periods 1–4 in market 1, 1–4 in market 2, 1–2 in market 3, 2–4 and 14 in market 4, and 1–3 in market 5. I leave out period 1 of market 4, in which six of the eight trades took place at 500 to 1,500 francs, far above any dividend; Plott and Sunder also exclude it from their means (p. 691). Period 14 of market 4 follows the insider periods, whereas all other no-information periods precede them. Appendix C repeats the comparisons with period 1 included, with the earlier and later periods separated, and with opening or closing prices in place of period means.

4.2 Five markets compared

Table 5 sets each market's downward-period prices beside its no-information price, and Figure 2 plots every period. With two to five periods per group, I emphasize magnitudes. The p-values, from Welch and rank-sum tests with periods as observations, are only a guide; Appendix C.3 reports inference with trades as observations.

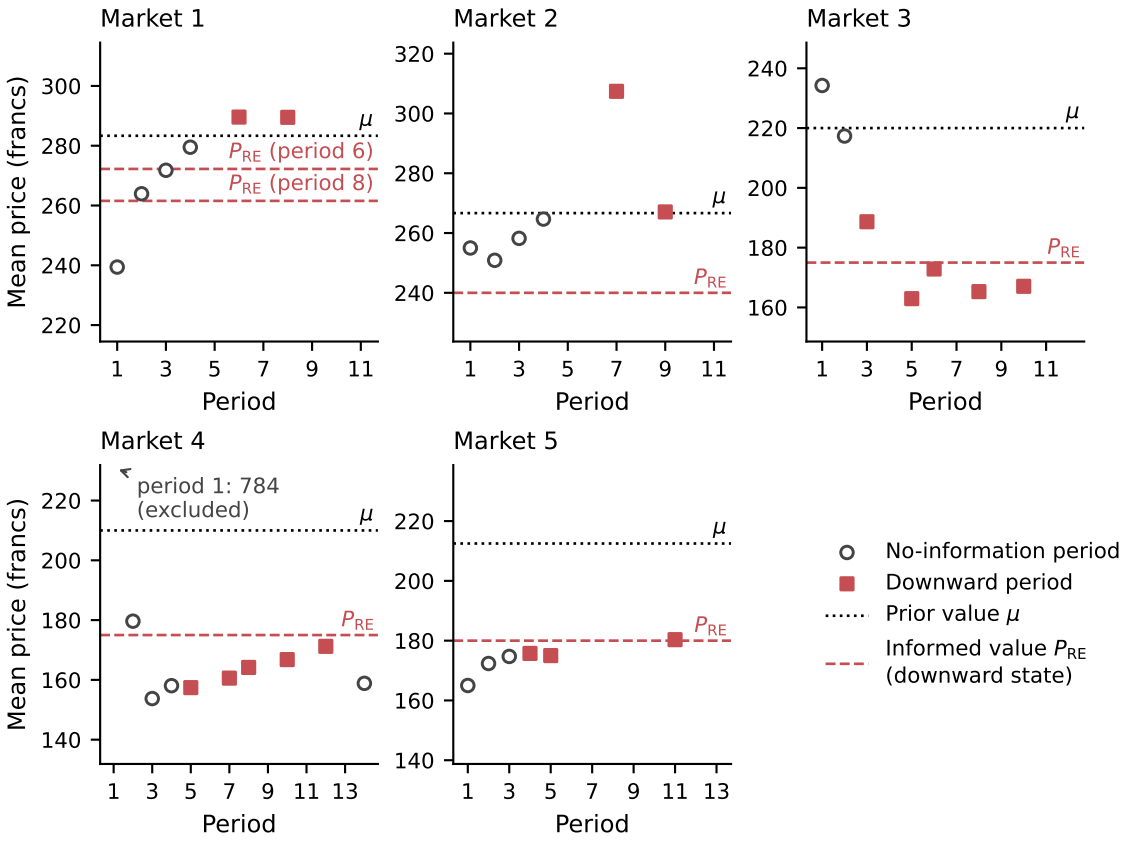


Figure 2: No-information and downward-period prices by market. Each marker is the mean transaction price of one period. Every panel spans 100 francs, so vertical distances are comparable across panels. The dotted line is the prior value μ and the dashed line the informed value P_{RE} in the downward state. Market 4's period 1 (mean 784 francs) lies off the scale and is excluded. Source: transaction rows in Appendix B of Plott and Sunder (1980); author's calculations.

Table 5: No-information and downward-period prices

			No-information		Downward			p-value	
Market	μ	P_{RE}	Periods	$\bar{p}_0$	Periods	Mean	Difference	Welch	Rank-sum
1	283.3	272.2, 261.5	4	263.7	2	289.5	+25.9	0.06	0.13
2	266.7	240.0	4	257.2	2	287.2	+30.0	0.37	0.13
3	220.0	175.0	2	225.8	5	171.4	−54.4	0.05	0.10
4	210.0	175.0	4	162.6	5	164.0	+1.5	0.83	0.56
5	212.5	180.0	3	170.7	3	177.0	+6.3	0.15	0.10

Notes: Prices are averages of period-mean transaction prices, in francs; $\bar{p}_0$ is the no-information price. P_{RE} is the informed value in the downward state; in market 1 it depends on the insiders' sample and differs between the two downward periods. Difference is the downward mean minus $\bar{p}_0$. The p-values are two-sided, from a Welch t-test and a Wilcoxon rank-sum test with periods as observations. Market 4's period 1 is excluded. Source: transaction rows in Appendix B of Plott and Sunder (1980); author's calculations.

The five markets fall into three groups. In market 3 (five downward periods), prices move from the no-information level toward RE. The no-information price, 226 francs, is near $\mu = 220$, and downward-period prices fall to 171, near the RE price of 175, a move of 54 francs ($p = 0.05$). This is what information entering prices looks like, and it supports the original conclusion.

In markets 4 and 5 (eight downward periods), prices were already there. No-information prices lie more than 40 francs below μ, at 163 against 210 in market 4 and 171 against 212.5 in market 5. They are near the RE price and almost equal to downward-period prices, which average 164 and 177. The downward-period evidence from these two markets cannot distinguish prices that incorporated insiders' information from prices that were already at that level.

In markets 1 and 2 (four downward periods), prices do not reach RE. Downward-period prices lie above μ and 26 to 30 francs above the no-information price. In market 1, mean prices drift upward over the whole session, from 239 francs in period 1 to 347 in period 11, and μ and RE are only 11 to 22 francs apart. In market 2, prices in period 9 fall to the RE level, 230 to 240 francs, in the second half of the period.

4.3 Interpretation

Plott and Sunder note the low no-information prices in markets 4 and 5 and suggest for market 4 that "no one knew that no one was informed" (pp. 685–686); in market 5, however, the absence of information was announced (p. 670). Other possibilities are that sellers were willing to sell below their prior values, that the uninformed type I traders, whose prior value sets μ, were unwilling to buy at μ, that traders were risk averse, and, in market 4, that prices fell back after the extreme prices of period 1. Appendix C.5 describes the subjects' experience in each market.

The original price test compares downward-period prices with the PI prediction, which equals μ (210 francs in market 4). In markets 4 and 5, however, the level that prices reach when no one can learn anything is the no-information price (163 in market 4), and it nearly coincides with the RE price (175). In these two markets, then, prices closer to RE than to PI do not show that information entered prices. Plott and Sunder observe that in these no-information periods prices "converge to an RE equilibrium price, but it is the wrong price, given the information" (p. 685); the same prices leave the downward-period comparison without a way to tell the two theories apart.

5 Price Paths in Upward Periods

Section 4 shows that, apart from market 3, downward-period prices do not show information entering prices. In upward periods, information has to move prices. How do they get to the informed value? The original article reports period-average prices, which hide the process; the trades listed in the working paper show it.

5.1 Data and empirical strategy

Appendix B of the working paper (Plott and Sunder, 1980, pp. 81–97) lists every bid, offer, and trade in sequence. I transcribed the 987 trades in all 61 periods; unaccepted bids and offers are not transcribed. The main sample is the 27 insider periods of markets 3–5, with 474 trades: 13 downward periods with 200 trades and

14 upward periods with 274 trades. These three markets have similar designs; in market 1 insiders see a noisy sample, and market 2 has all-informed training periods before its insider periods. Appendix Table B2 lists the 27 periods.[2]

I summarize where a price p lies by its *price position*

$$D(p) = \frac{p - \mu}{P_{\mathrm{RE}} - \mu}. \tag{4}$$

D equals 0 at the prior value and 1 at the informed value. The denominator is signed, which puts adjustment in upward and downward periods on the same scale: in both, D moving from 0 toward 1 means the price moving from the prior value toward the informed value. Figure 3 shows the scale for the two states of market 4. $D < 0$ means the price lies on the far side of the prior value, and $D > 1$ means it has passed the informed value. D_1, $\bar{D}$, and D_T apply equation (4) to the first trade, the period-mean price, and the last trade of a period. In the path figures, the kth of N trades in a period is placed at $u = (k-1)/(N-1)$, with $u = 0$ when $N = 1$. The scale orders trades; it does not measure time, which the appendix does not record.

I compare the two kinds of period at the opening, during the period, and at the close. Because D divides by the distance from μ to P_{RE}, which is shorter in downward periods, I also report deviations from the informed value in francs. The signed deviation shows on which side of P_{RE} a price lies, and the absolute deviation shows how far it is from P_{RE}, unaffected by the shorter distance in downward periods. Each period receives equal weight.

[2]In 58 of the 61 periods, the mean of the transcribed prices rounds to the integer printed in the original, and in two more it lies about 0.02 francs beyond the rounding boundary. The exception is market 1, period 2: the 14 trades listed in the appendix, rechecked against the page image, average 263.9 francs, while the original prints 269; the difference appears to come from the original's averaging. The period enters only market 1's no-information price in Section 4; replacing its mean with 269 raises that price by about 1 franc and the Welch p-value in Table 5 from 0.06 to 0.07, and leaves the rank-sum p-value unchanged (Appendix B.1).

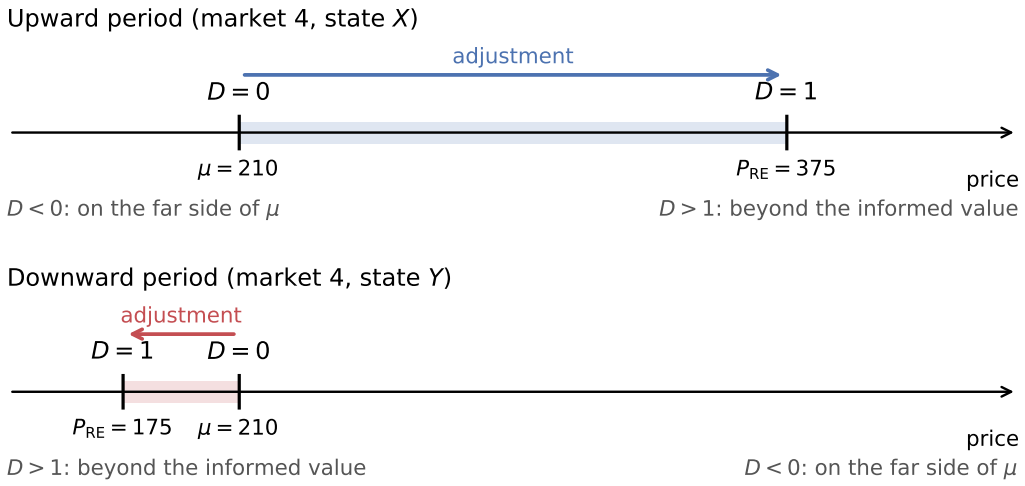


Figure 3: The price position D in market 4. In both states, D runs from 0 at the prior value μ to 1 at the informed value P_{RE} (shaded). The arrows point in the direction of adjustment.

5.2 Results

Opening prices differ sharply (Table 6). In upward periods the first trade averages 83 francs below the informed value, and D_1 averages 0.30: the first trade covers about 30 percent of the distance from the prior value to the informed value. All 14 upward periods open below P_{RE}, and 4 of them (29 percent) open on the far side of the prior value ($D_1 < 0$). In downward periods the first trade averages 3 francs below P_{RE} and 15 francs away from it in absolute value; 8 of the 13 periods open below P_{RE}, 2 exactly at it, and 3 above it.

Prices then rise in upward periods. From the first to the last trade they rise by 71 francs on average (median 59); 11 of the 14 upward periods rise, and 3 end where they began. In downward periods the average change is -2 francs (median 0). Figure 4 shows the paths.

By the close, the franc gap is small. Upward periods end 12 francs below P_{RE} on average and downward periods 5 francs below, a difference that is not statistically significant ($p = 0.19$); both medians are -5 francs. Positions still differ: D_T averages

Table 6: Price positions at the opening, over the period, and at the close

Period type	Periods	Trades	D_1	$\bar{D}$	D_T	Opening $p_1 - P_{\mathrm{RE}}$ Signed	Opening $p_1 - P_{\mathrm{RE}}$ Absolute	$p_T - p_1$
Downward	13	200	1.09	1.17	1.15	−3.1 (−5.0)	15.4 (15.0)	−2.3 (0.0)
Upward	14	274	0.30	0.67	0.89	−82.5 (−70.0)	82.5 (70.0)	70.8 (59.0)

Notes: The sample is the 27 insider periods of markets 3–5. $D = (p - \mu)/(P_{\mathrm{RE}} - \mu)$ equals 0 at the prior value and 1 at the informed value. D_1, $\bar{D}$, and D_T use the first trade, the period-mean price, and the last trade; periods receive equal weight. Franc columns give means with medians in parentheses: the signed and absolute deviation of the first trade from P_{RE}, and the change from the first to the last trade. Source: transaction rows in Appendix B of Plott and Sunder (1980); author's calculations.

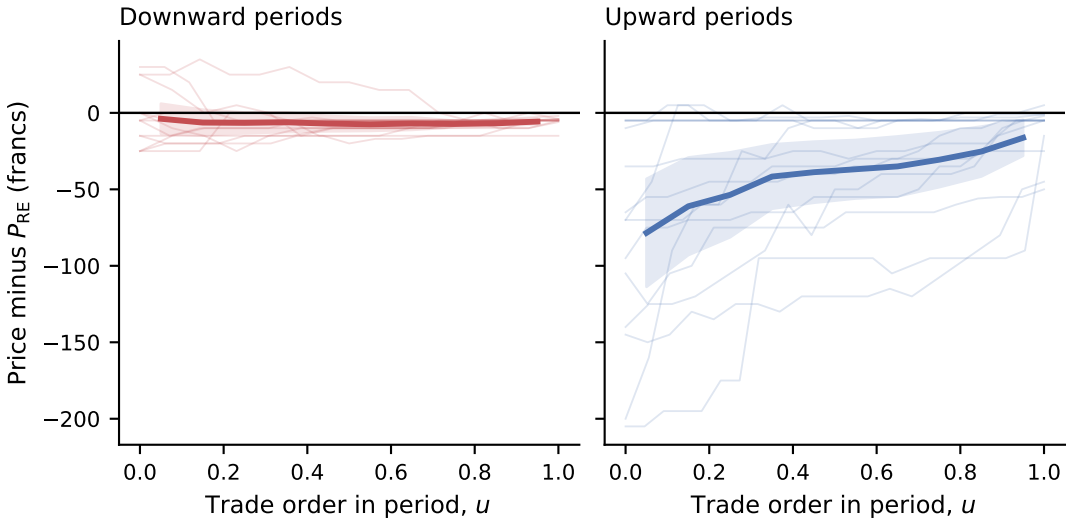


Figure 4: Transaction prices within downward and upward periods. The vertical axis is price minus the informed value P_{RE}, in francs. Thin lines trace individual periods by trade order $u = (k - 1)/(N - 1)$. Thick lines are means over ten equal-width bins of u, averaged first within period and bin and then across periods, plotted at the bin centers; bands are 95 percent intervals. Because they average over bins, the first and last points differ from the opening and closing prices in Table 6. The sample is 13 downward and 14 upward periods in markets 3–5, with 474 trades. Source: transaction rows in Appendix B of Plott and Sunder (1980); author's calculations.

0.89 in upward periods and 1.15 in downward periods ($p < 0.001$), and 13 of the 14 upward periods close below P_{RE}. Closing prices alone would hide the differences at the opening and during the period. In downward periods D_1 and D_T are slightly above 1, so prices end slightly below P_{RE}, beyond the informed value; this matches Section 4, where prices in markets 4 and 5 were near or below P_{RE} even when no one was informed.

Plott and Sunder exempt "a new and unknown state of nature" from their claim that prices adjust instantaneously (p. 685). Restricting the sample to states that had already appeared in at least two earlier insider periods, the opening position averages 0.99 in 7 downward periods and 0.72 in 6 upward periods. The upward mean remains below 1 ($p = 0.017$), but the difference between the two kinds of period is no longer precisely estimated ($p = 0.28$). Appendix D reports further checks. They include an opening window of the first 10 percent of trades, permutation tests within markets, and market fixed effects. Pooling all five markets, upward periods still open 69 francs below P_{RE} on average, although markets 1 and 2 show little adjustment within periods. The appendix also reports a closeness index and measures positions from the no-information price instead of μ.

6 An LLM Pretest

Why do upward periods adjust slowly? Sections 3.3 and 5 leave the question open, because distance, insider agreement, and uninformed competition all change with the direction of adjustment in the original design. Separating them requires a new human experiment. This section pretests such an experiment with markets in which every trader is an LLM agent. Its purpose is to choose designs and treatments for the human experiment, and its results are not evidence about human behavior.

6.1 Implementation

The LLM markets keep the original endowments, the cash and inventory constraints, a public book with one standing bid and one standing offer, and one certificate per

trade. They differ from the original in five ways: traders act in random order, normally at most three times per period; a new bid or offer must improve the standing one; traders see only the current period's public record; their private memory is limited; and every action comes with an elicited belief about the state and a reservation price. Most sessions use DeepSeek-V4-Flash at temperature 0.7 with reasoning enabled, and one market-3 session uses a second model, Gemini 3.5 Flash. Appendix Table E1 compares the institutions, and Appendix Table E2 lists every session.

6.2 Results with the original parameters

Across 26 sessions of markets 1–5, with 188 traded insider periods, LLM prices reach and pass the informed value in downward periods: the period-mean position averages 1.10 and the closing position 1.27 (Table 7). In upward periods they stay at or below the prior value, at -0.09 on average and -0.14 at the close. Market 4, the reference for the symmetric markets below, shows the same pattern in its five sessions: the period-mean position is 1.29 in downward periods and 0.11 in upward periods, which close at 0.22. Human prices in market 4's upward periods move from 0.29 at the opening to 0.89 at the close (Appendix Table D1), an adjustment the LLM markets do not reproduce. With the original parameters, then, the LLM markets reproduce downward periods but not the human adjustment in upward periods, and giving traders more turns per period does not reliably narrow the gap (Appendix E.5).

6.3 Symmetric markets A and B

Markets A and B remove the three differences of Section 3.3 (Table 8). In both directions the informed value lies 100 francs from the prior value. At the prior value, all six insiders trade in the same direction, buying in upward periods and selling in downward periods. In neither direction can an uninformed trader trade at the target price without learning the state. The two markets differ in one respect: in A, the same type sets the prior value and the upward target, while in B different types do.

Two features cannot be removed by changing parameters: the RE and PI predictions are equal in upward periods (Proposition 1), and the trading rules give the

Table 7: Price positions in LLM markets with original and symmetric parameters

Sample	Period type	Sessions	Periods	D_1	$\bar{D}$	D_T	$D_T - D_1$
Market 4	Upward	5	19	−0.07	0.11	0.22	+0.30
Market 4	Downward	5	26	1.31	1.29	1.38	+0.07
Markets A and B	Upward	6	33	0.17	0.41	0.49	+0.33
Markets A and B	Downward	6	20	0.21	0.40	0.76	+0.55
Markets 1–5	Upward	26	99	−0.20	−0.09	−0.14	+0.06
Markets 1–5	Downward	26	89	1.22	1.10	1.27	+0.05

Notes: $D = (p - \mu)/(P_{\mathrm{RE}} - \mu)$ equals 0 at the prior value and 1 at the informed value. D_1, $\bar{D}$, and D_T use the first trade, the period-mean price, and the last trade. Entries give traded insider periods equal weight. Markets A and B use seeds 42–44 in each market. Market 4 and markets A and B differ jointly in dividends, prior, and the conditions in Table 8, and their sessions are not matched on seed or state sequence.

Table 8: Differences between upward and downward periods: markets 3–5 and the symmetric markets

	Distance $P_{\mathrm{RE}} - \mu$		Insiders buying minus selling at μ		Uninformed trading at P_{RE} at prior value	
Market	Upward	Downward	Upward	Downward	Upward	Downward
3	+180	−45	+2	−6	0	2
4	+165	−35	+2	−6	0	2
5	+32.5 or +107.5	−32.5	−2	−6	0	4
A	+100	−100	+6	−6	0	0
B	+100	−100	+6	−6	0	0

Notes: Columns as in Table 4. Markets A and B keep market 4's population, endowments, information schedule, and number of periods; their dividends and prior are in Appendix Tables A1 and A2. Source: author's calculations.

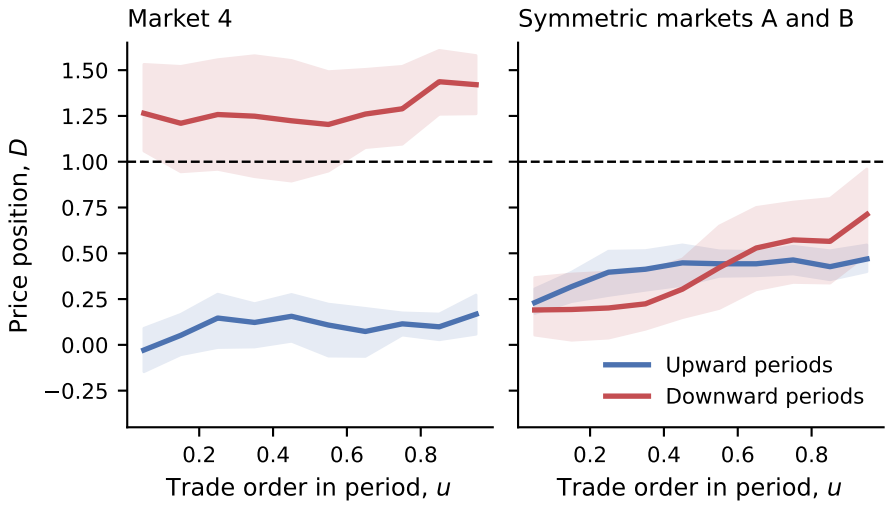


Figure 5: Price positions within periods in LLM markets. $D = 0$ is the prior value and $D = 1$ (dashed) the informed value. The horizontal axis is trade order $u = (k-1)/(N-1)$, with ten equal-width bins plotted at their centers. Means average trades within period and bin, then periods within session and bin, then sessions equally; bands resample whole sessions for 95 percent intervals. The left panel uses five market-4 sessions and the right panel three seeds in each of markets A and B.

two sides different rents and quantities, since sellers can sell at most two certificates while buyers have ample capital. Markets A and B also change dividends and the prior, raising $\pi(X)$ from 0.4 to 0.6, so the comparison with market 4 tests the joint effect of these changes and cannot isolate any single difference.

The directional gap largely disappears (Table 7 and Figure 5). The difference between downward and upward period-mean positions falls from 1.18 in market 4 to -0.01 in markets A and B. Both directions, however, reach only about 0.4: 0.41 in upward periods and 0.40 in downward periods. Closing positions still differ, at 0.49 and 0.76. Weighting sessions equally, the directional gap is -0.02, and individual sessions have gaps of both signs (Appendix E.3).

6.4 Step-by-step disclosure

Starting from the baseline of markets A and B, I tell traders more about the market in four steps (Table 9). Step 1b discloses nothing new; it bundles three changes to how the market is implemented. Each step keeps the dividends, prior, endowments, and allocation of clues of its market and uses the state sequence of its seed-matched baseline. Appendix E.1 gives the treatment text.

Table 9: Steps of disclosure in markets A and B

Step	Added to the previous step
0	Baseline: traders know only their own dividends; they are not told how many traders, or which ones, receive clues.
1	All three dividend schedules are public, and traders are told that two traders of each type receive a clue in a period with clues.
1b	Three implementation changes: an explicit objective of earning francs, stronger emphasis that a clue is always correct, and one persistent private memo in place of rolling notes.
2	Each period, traders are told whether anyone received a clue in that period.
3	Traders are told that the same traders receive the clues in every period with clues, without being told who they are.

Over the full sequence, prices move toward the informed value. From step 0 to step 3, in four session pairs matched by market, seed, period, and period type, the upward-period position rises from 0.31 to 0.82 and rises in all four pairs; the downward-period position rises from 0.47 to 0.73 and rises in three of the four. Single steps are less consistent. Disclosing the dividend schedules moves the downward-period position in opposite directions in the two markets (-0.31 in A and $+0.32$ in B). Disclosing that the same traders always receive the clues raises the upward-period position but lowers the downward-period position on average. Figure 6 traces the five steps at seed 42, the only seed with all five steps in both markets. As prices move toward the informed value, uninformed traders' average belief in the realized state rises too, from 0.58 to 0.82 in downward periods between steps 1b and 2 and from 0.86 to 0.91 in upward periods between steps 2 and 3, consistent with learning from prices. Appendix E.4 gives the paired estimates, the results by session pair, a common-support check, beliefs and allocations, and the tests.

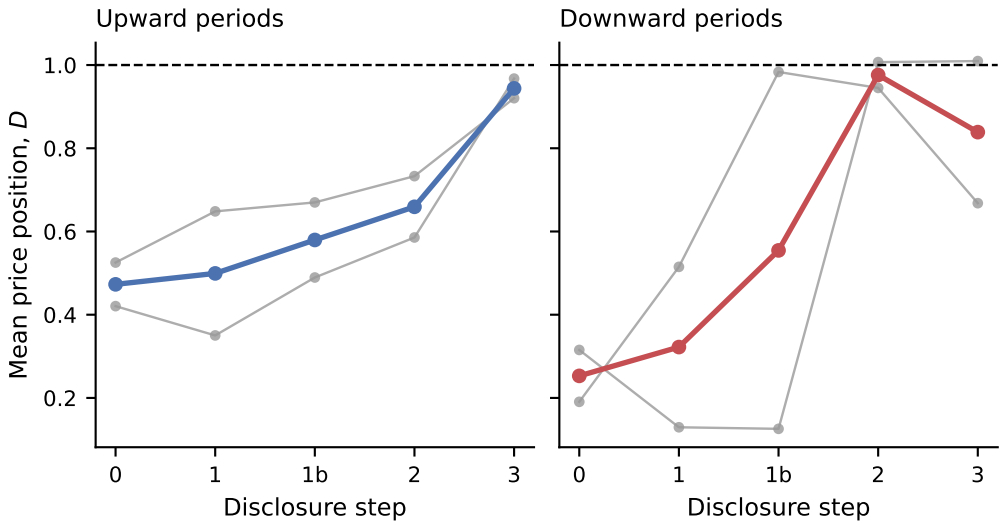


Figure 6: Price positions across the disclosure steps at seed 42. Thin gray lines are the means for markets A and B; the colored line weights the two markets equally. $D = 0$ is the prior value and $D = 1$ (dashed) the informed value. Steps are defined in Table 9. Samples and weights differ from the matched estimates in the text.

For a human experiment, the pretest suggests two treatments worth testing. The first is a design that is symmetric between the two directions and in which upward periods can also distinguish RE from PI. By Proposition 1, parameters alone cannot achieve this under the original rules, so the design must change who is informed, for example by giving no insider to the type with the highest dividend in an upward state (Section 3.2). The second is public disclosure of whether insiders are present and that they are always the same traders. It also suggests what to record: time stamps of bids, offers, and trades; traders' beliefs during trading; and periods without trade.

7 Conclusion

The periods that can test the two theories mostly cannot show whether information entered prices, and in the periods in which information has to move prices, prices adjust gradually. The original price conclusion rests on 17 downward periods in five markets. Only in market 3 do prices move from their no-information level toward

RE. In markets 4 and 5, prices were already near RE when no one was informed; in markets 1 and 2, they stayed above RE. In upward periods, prices open far below the informed value and do most of their adjusting while trading continues.

These findings bear on how such experiments are designed and read. A price test discriminates between two theories only if their predictions differ and prices do not settle at the same place without any information, so experiments of this kind should report a no-information benchmark. Period-average and closing prices hide the adjustment process, while transaction paths show how information enters prices. Designs should let both directions distinguish the theories. The condition that matters most is whether the type with the highest dividend in an upward state includes insiders (Proposition 1). The distance prices must travel, the agreement among insiders, and the competition from uninformed traders should also be symmetric (Sections 3.3 and 6), and the asymmetry built into the rules, which ban short sales and give buyers ample capital, should be kept in view. Experimental evidence cited for the claim that prices transmit insiders' information should be read with these limits in mind.

The evidence has limits of its own. Each market has only a few no-information and downward periods, so inference rests mainly on magnitudes. The comparison of downward and upward periods is descriptive, because the direction of adjustment comes bundled with the other differences of Section 3.3. The next step is a human experiment that is symmetric between buying and selling, in which both directions can distinguish RE from PI, and that records when bids and offers are made and what traders believe during trading.

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A Parameters and Theory

A.1 Market parameters

All markets reset each trader to two certificates and 10,000 francs at the start of a period and subtract a fixed cost of 10,000 francs at its end. Markets 1 and 2 convert 500 francs to one dollar, and the other markets convert 333.33 francs to one dollar. Table A1 lists the priors, populations, information schedules, and exchange rates of the five original markets and of the symmetric markets A and B used in Section 6. "Informed" is the number of traders who receive a clue in an insider period: two of each type in markets 2–5, A, and B, and one of each type in market 1.

Table A1: Market structure, priors, and information schedules

Market	Traders	Informed	States	Prior	Periods	Information schedule	Francs/$
1	9	3	X/Y	X:0.33, Y:0.67	11	none 1–4; insider 5–8; all 9–11	500
2	12	6	X/Y	X:0.33, Y:0.67	11	none 1–4; all 5–6; insider 7–11	500
3	12	6	X/Y	X:0.40, Y:0.60	12	none 1–2; insider 3–10; all 11–12	333
4	12	6	X/Y	X:0.40, Y:0.60	14	none 1–4; insider 5–13; none 14	333
5	12	6	X/Y/Z	X:0.35, Y:0.25, Z:0.40	13	none 1–3; insider 4–13	333
A	12	6	X/Y	X:0.60, Y:0.40	14	none 1–4; insider 5–13; none 14	333
B	12	6	X/Y	X:0.60, Y:0.40	14	none 1–4; insider 5–13; none 14	333

A.2 Dividend schedules

Table A2 gives every dividend used in the calculations. Markets A and B keep market 4's population, endowments, information schedule, and number of periods, and change its dividends and prior: $\pi(X)$ is 0.6 in A and B and 0.4 in market 4.

Table A2: Dividend schedules (francs per certificate)

Market	Type	State X	State Y	State Z
1	I	150	350	–
1	II	250	300	–
1	III	300	100	–
2	I	100	350	–
2	II	200	300	–
2	III	240	175	–
3	I	400	100	–
3	II	300	150	–

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Continued

Market	Type	State X	State Y	State Z
3	III	125	175	–
4	I	375	100	–
4	II	275	150	–
4	III	100	175	–
5	I	120	170	320
5	II	155	245	135
5	III	180	100	160
A	I	360	110	–
A	II	330	130	–
A	III	290	160	–
B	I	380	180	–
B	II	350	200	–
B	III	400	100	–

A.3 Prior values, informed values, and PI predictions

For markets with exact clues, μ is the highest prior expected dividend, $P_{\text{RE}}(\theta)$ the highest realized dividend, and $P_{\text{PI}}(\theta) = \max\{P_{\text{RE}}(\theta), \mu\}$. Table A3 gives the values by market and state, with the period type each state produces in an insider period. Market 1 is in Appendix A.4.

Table A3: Prior value, informed value, and PI prediction by market and state

Market	State	μ	P_{RE}	P_{PI}	$P_{\text{RE}} - \mu$	Period type
2	X	266.7	240.0	266.7	-26.7	Downward
2	Y	266.7	350.0	350.0	+83.3	Upward
3	X	220.0	400.0	400.0	+180.0	Upward
3	Y	220.0	175.0	220.0	-45.0	Downward
4	X	210.0	375.0	375.0	+165.0	Upward
4	Y	210.0	175.0	210.0	-35.0	Downward
5	X	212.5	180.0	212.5	-32.5	Downward
5	Y	212.5	245.0	245.0	+32.5	Upward
5	Z	212.5	320.0	320.0	+107.5	Upward
A	X	260.0	360.0	360.0	+100.0	Upward
A	Y	260.0	160.0	260.0	-100.0	Downward
B	X	300.0	400.0	400.0	+100.0	Upward
B	Y	300.0	200.0	300.0	-100.0	Downward

A.4 Market 1's probabilistic clue

In market 1, an insider observes a clue c, ten draws with replacement from one of two boxes, rather than the state. Conditional on state X a draw equals one with probability 0.2, and conditional on Y with probability 0.4. With prior probabilities 1/3 and 2/3, a clue containing k ones gives

$$q(X \mid k) = \frac{\frac{1}{3}(0.2)^k(0.8)^{10-k}}{\frac{1}{3}(0.2)^k(0.8)^{10-k} + \frac{2}{3}(0.4)^k(0.6)^{10-k}},$$

and the clue-conditional value of type t is $d_t(c) = q(X \mid k)\, d_t(X) + [1 - q(X \mid k)]\, d_t(Y)$. Full revelation then means that prices reveal the clue, which leaves some uncertainty about the state.

Corollary 1. *Suppose insiders observe c rather than θ and that, under RE, the price reveals c. With (A2)–(A4) otherwise unchanged, Proposition 1 holds with $d_t(c)$ in place of $d_t(\theta)$:*

$$P_{\text{RE}}(c) = \max_t d_t(c), \qquad P_{\text{PI}}(c) = \max\{P_{\text{RE}}(c), \mu\}.$$

Proof. The proof of Proposition 1 uses the state only through the value that an insider, and under RE every trader, attaches to a certificate. For a risk-neutral

trader who observes c, that value is $d_t(c)$, while uninformed PI values remain the prior expected dividends. Every step goes through with $d_t(c)$ in place of $d_t(\theta)$. □

Table A4 shows the four insider periods. Two clues put the clue-conditional RE price above $\mu = 283.3$ and make the period upward; two put it below and make the period downward. Table 3 and Figure 1 use these values.

Table A4: Market 1: clue-conditional benchmarks in the insider periods

Period	State	Ten-draw clue	Ones	$q(X \mid c)$	$P_{\mathrm{RE}}(c)$	$P_{\mathrm{PI}}(c)$	Period type	Differ?
5	Y	0100101010	4	0.1494	320.128	320.128	Upward	No
6	X	0000000011	2	0.5553	272.236	283.333	Downward	Yes
7	Y	0100110100	4	0.1494	320.128	320.128	Upward	No
8	Y	0000010000	1	0.7690	261.548	283.333	Downward	Yes

A.5 Allocation predictions and the scope of the other original tests

The original efficiency comparison concentrates on the periods in which the RE and PI price predictions differ (Plott and Sunder, 1982, p. 685). These are the 17 downward periods; the original Table 4 displays 15 of them, omitting market 1. The allocation comparison in the original Table 5 covers all 36 insider periods, and the profit comparison in Table 6 covers both kinds of period (Plott and Sunder, 1982, pp. 683–685).

A sharper allocation comparison concerns periods in which the two theories predict disjoint sets of holders. The values that set the two prices also determine who holds the certificates, and Lemma 1 states when the predicted sets are nested and when they are disjoint.

Lemma 1. *Suppose (A1)–(A4) hold in an insider period with state θ. Let t^* be the unique type with the highest realized dividend, and let $\bar{T}$ be the set of types whose prior expected dividend equals μ.*

(a) In an upward period, PI predicts that the insiders of type t^* hold the certificates and RE that all members of t^* do. The PI set is contained in the RE set.

(b) In a downward period, PI predicts that the uninformed members of the types in $\bar{T}$ hold the certificates and RE that the members of t^* do. The two sets are disjoint if and only if $t^* \notin \bar{T}$.

Proof. Under (A4), the certificates go to the traders with the highest value, and any one of them can absorb the whole supply. Under RE, every member of t^* values a certificate at $P_{\mathrm{RE}}(\theta)$, and every other trader at a lower realized dividend, because t^* is unique. This gives the RE set in both parts. (a) Under PI, the insiders of t^*, who exist by (A3), value a certificate at $P_{\mathrm{RE}}(\theta)$. Every other insider values it at some $d_t(\theta) < P_{\mathrm{RE}}(\theta)$, and every uninformed trader at a prior expected dividend no greater than $\mu < P_{\mathrm{RE}}(\theta)$. The PI holders are therefore the insiders of t^*, a subset of the RE holders. (b) Under PI, the uninformed members of the types in $\bar{T}$, who exist by (A3), value a certificate at μ. Every other uninformed trader values it below μ, and every insider at some $d_t(\theta) \leq P_{\mathrm{RE}}(\theta) < \mu$. A trader belongs to both sets only if the trader is an uninformed member of t^* and $t^* \in \bar{T}$; conversely, if $t^* \in \bar{T}$, the uninformed members of t^* belong to both. □

Part (a) explains why an upward period cannot produce disjoint predictions. Part (b) makes disjointness in a downward period a property of the dividends. In each downward state of markets 2–5, the highest-dividend type is type III, while the highest prior value belongs to type I, together with type II in market 2 (Table A2), so the predicted sets are disjoint.

Market 1 is the exception. With the clue-conditional values of Corollary 1, types I and II tie for the highest prior value, and type II has the highest clue-conditional value in both downward periods, so $t^* \in \bar{T}$. PI assigns the certificates to the uninformed members of types I and II, RE assigns them to type II, and the two sets share the uninformed type II traders. The original Table 5 nevertheless marks both periods as disjoint (Appendix B.3, item 4). The classification of downward and upward periods is unaffected.

B Data Transcription and the Original Record

B.1 Transcription and period-by-period checks

The human data are a transcription of the transaction rows of Appendix B, "Bids, Offers, Prices" (pp. 81–97), of the working paper (Plott and Sunder, 1980). The appendix lists bids, offers, and trades in one numbered sequence. A row enters the transaction file only when the appendix lists a buyer, a seller, and a numeric price, so unaccepted bids and offers are not transcribed. The file contains 987 trades in all 61 periods. I match each trade to its market, period, state, information condition, and price levels, and compare each period's transcribed mean with the integer mean printed in the original's figures. Table B1 reports all 61 comparisons.

In 58 periods the printed integer is the transcribed mean rounded to the nearest franc, and in two more the transcribed mean lies only 0.022 or 0.024 francs beyond the rounding boundary. The exception is market 1, period 2. Its 14 listed trades, which I rechecked against the page image, are 250, 255, 265, 260, 255, 260, 260, 275, 275, 275, 260, 275, 270, and 260 francs; they average 263.93, whereas the original prints 269. Reaching 269 would require one more trade at 340 francs, far outside the plotted range, so the printed mean appears to be an arithmetic slip. The period is a no-information period and enters only market 1's no-information price in Section 4; replacing its mean with 269 raises that price from 263.7 to 264.9 francs and market 1's Welch p-value in Table 5 from 0.059 to 0.068; the rank-sum p-value is unchanged.

Table B1: Period-by-period audit of the 987 human transactions

Market	Period	Trades	Printed mean	Transcribed mean	Error	Check
1	1	17	239.0	239.412	+0.412	ok
1	2	14	269.0	263.929	-5.071	mean_mismatch
1	3	14	272.0	271.786	-0.214	ok
1	4	10	280.0	279.500	-0.500	ok
1	5	17	289.0	289.412	+0.412	ok
1	6	12	290.0	289.583	-0.417	ok
1	7	14	301.0	300.714	-0.286	ok
1	8	10	289.0	289.500	+0.500	ok
1	9	16	310.0	309.688	-0.312	ok
1	10	17	328.0	327.941	-0.059	ok
1	11	19	347.0	346.579	-0.421	ok

Continued on next page

Continued

Market	Period	Trades	Printed mean	Transcribed mean	Error	Check
2	1	11	255.0	255.000	+0.000	ok
2	2	8	251.0	250.875	-0.125	ok
2	3	14	258.0	258.214	+0.214	ok
2	4	12	265.0	264.667	-0.333	ok
2	5	20	304.0	303.500	-0.500	ok
2	6	16	337.0	336.938	-0.062	ok
2	7	22	307.0	307.409	+0.409	ok
2	8	19	328.0	328.368	+0.368	ok
2	9	17	267.0	267.059	+0.059	ok
2	10	19	327.0	327.316	+0.316	ok
2	11	20	335.0	334.800	-0.200	ok
3	1	12	234.0	234.250	+0.250	ok
3	2	15	217.0	217.333	+0.333	ok
3	3	15	189.0	188.667	-0.333	ok
3	4	19	333.0	332.895	-0.105	ok
3	5	15	163.0	162.933	-0.067	ok
3	6	14	173.0	172.857	-0.143	ok
3	7	19	364.0	364.053	+0.053	ok
3	8	14	165.0	165.286	+0.286	ok
3	9	16	396.0	395.875	-0.125	ok
3	10	15	167.0	167.067	+0.067	ok
3	11	18	177.0	176.778	-0.222	ok
3	12	18	395.0	394.722	-0.278	ok
4	1	8	784.0	784.375	+0.375	ok
4	2	14	180.0	179.643	-0.357	ok
4	3	16	154.0	153.750	-0.250	ok
4	4	16	158.0	158.062	+0.062	ok
4	5	17	157.0	157.412	+0.412	ok
4	6	23	254.0	253.478	-0.522	ok
4	7	17	161.0	160.588	-0.412	ok
4	8	12	164.0	164.167	+0.167	ok
4	9	20	301.0	300.700	-0.300	ok
4	10	14	167.0	166.786	-0.214	ok
4	11	21	323.0	323.095	+0.095	ok
4	12	18	171.0	171.222	+0.222	ok
4	13	21	344.0	343.476	-0.524	ok
4	14	13	159.0	158.846	-0.154	ok
5	1	10	165.0	165.000	+0.000	ok
5	2	17	172.0	172.353	+0.353	ok
5	3	18	175.0	174.722	-0.278	ok
5	4	15	176.0	175.733	-0.267	ok
5	5	16	175.0	175.000	+0.000	ok
5	6	17	237.0	237.059	+0.059	ok
5	7	20	206.0	206.250	+0.250	ok
5	8	23	277.0	277.391	+0.391	ok
5	9	21	240.0	240.000	+0.000	ok
5	10	20	240.0	240.000	+0.000	ok
5	11	18	180.0	180.333	+0.333	ok
5	12	18	240.0	240.056	+0.056	ok
5	13	16	296.0	295.625	-0.375	ok

B.2 The main sample

Table B2 lists the 27 insider periods of markets 3–5 used in Section 5, with the state, period type, number of trades, price levels, opening, mean, and closing prices, and the three price positions. The rows sum to 474 trades.

Table B2: The 27 human insider periods in the main sample

Mkt.	Per.	State	Type	N	μ	P_{RE}	p_1	$\bar{p}$	p_T	D_1	$\bar{D}$	D_T
3	3	Y	Downward	15	220.0	175.0	200.0	188.7	170.0	0.444	0.696	1.111
3	4	X	Upward	19	220.0	400.0	260.0	332.9	395.0	0.222	0.627	0.972
3	5	Y	Downward	15	220.0	175.0	150.0	162.9	169.0	1.556	1.268	1.133
3	6	Y	Downward	14	220.0	175.0	200.0	172.9	170.0	0.444	1.048	1.111
3	7	X	Upward	19	220.0	400.0	200.0	364.1	399.0	-0.111	0.800	0.994
3	8	Y	Downward	14	220.0	175.0	175.0	165.3	171.0	1.000	1.216	1.089
3	9	X	Upward	16	220.0	400.0	390.0	395.9	399.0	0.944	0.977	0.994
3	10	Y	Downward	15	220.0	175.0	170.0	167.1	173.0	1.111	1.176	1.044
4	5	Y	Downward	17	210.0	175.0	150.0	157.4	160.0	1.714	1.503	1.429
4	6	X	Upward	23	210.0	375.0	170.0	253.5	360.0	-0.242	0.264	0.909
4	7	Y	Downward	17	210.0	175.0	160.0	160.6	170.0	1.429	1.412	1.143
4	8	Y	Downward	12	210.0	175.0	160.0	164.2	170.0	1.429	1.310	1.143
4	9	X	Upward	20	210.0	375.0	270.0	300.7	325.0	0.364	0.550	0.697
4	10	Y	Downward	14	210.0	175.0	150.0	166.8	170.0	1.714	1.235	1.143
4	11	X	Upward	21	210.0	375.0	280.0	323.1	370.0	0.424	0.685	0.970
4	12	Y	Downward	18	210.0	175.0	170.0	171.2	170.0	1.143	1.108	1.143
4	13	X	Upward	21	210.0	375.0	310.0	343.5	373.0	0.606	0.809	0.988
5	4	X	Downward	15	212.5	180.0	180.0	175.7	176.0	1.000	1.131	1.123
5	5	X	Downward	16	212.5	180.0	175.0	175.0	175.0	1.154	1.154	1.154
5	6	Y	Upward	17	212.5	245.0	175.0	237.1	250.0	-1.154	0.756	1.154
5	7	Z	Upward	20	212.5	320.0	175.0	206.2	275.0	-0.349	-0.058	0.581
5	8	Z	Upward	23	212.5	320.0	250.0	277.4	295.0	0.349	0.604	0.767
5	9	Y	Upward	21	212.5	245.0	240.0	240.0	240.0	0.846	0.846	0.846
5	10	Y	Upward	20	212.5	245.0	240.0	240.0	240.0	0.846	0.846	0.846
5	11	X	Downward	18	212.5	180.0	210.0	180.3	176.0	0.077	0.990	1.123
5	12	Y	Upward	18	212.5	245.0	240.0	240.1	240.0	0.846	0.848	0.846
5	13	Z	Upward	16	212.5	320.0	285.0	295.6	315.0	0.674	0.773	0.953

B.3 Five inconsistencies in the original tables and figures

Rebuilding the period data exposes five places where the published tables and figures of Plott and Sunder (1982) disagree with one another (Table B3). All five appear in the same form in the working paper (Plott and Sunder, 1980). None affects the original conclusions or any sample or estimate in this paper, but items 1, 2, 3, and 5 matter for anyone rebuilding the period data. Unqualified table, figure, and page numbers refer to the published article.

Table B3: Inconsistencies in the original tables and figures

Item	Inconsistency	Published article	Working paper
1	Market 3, period 1: state	Table 1 gives Y; Figure 4 and Tables 5–6 give X. Appears to be X.	Figure 1 gives Y; Figures 9, 14, and 16 give X.
2	Market 5, period 1: state	Figure 6 gives Y; Tables 1, 5, and 6 give Z. Appears to be Z.	Figure 11 gives Y; Figures 1, 14, and 16 give Z.
3	Market 3 periods on the 17-period list	Footnote 6 lists 3, 5, 7, 8, 10; Table 5's asterisks fall on 3, 5, 6, 8, 10. Appears to be 6, not 7.	Footnote 3 (p. 98) lists 7; Figures 12 and 17 list 6.
4	Market 1, periods 6 and 8: disjoint predictions	Table 5's note and p. 684 cover all 17 asterisked periods; Table 3 covers only markets 2–5. The predicted holder sets overlap in these two periods.	p. 34 states 17 disjoint periods; Figure 17 lists 15, from markets 2–5 only.
5	Market 4, periods 7 and 8: plotted trades	Trades plotted in Figure 5 lie about 10 francs above the printed averages, 161 and 164.	Figure 10 shows the same offset; Appendix B gives 160.59 and 164.17.

Notes: Working-paper numbering differs from the article: published Table 1 is its Figure 1, Table 2 its Figure 2, Table 3 its Figures 4–5, Table 4 its Figure 17, Table 5 its Figure 14, Table 6 its Figure 16, Figures 2–6 its Figures 7–11, and footnote 6 its footnote 3. Its Figures 6 and 12 and Appendix B have no published counterpart.

1. Market 3, period 1: the state. Table 1 gives Y; three other places give X. Figure 4 labels the first panel 1(X), and the realized-state row for market 3 in Table 5, X Y Y X Y Y X Y X Y Y X, matches Figure 4 in all twelve periods. Table 6 agrees: its X row has five entries and its Y row seven, the split Figure 4 implies,

and the period-1 entry, 76, sits in the X row. The working paper shows the same split: its design table (Figure 1, p. 9) gives Y, while its Figures 9, 14, and 16 give X. Period 1 informs nobody, so the discrepancy enters neither the classification of insider periods nor any comparison here.

2. Market 5, period 1: the state. Figure 6 labels the first panel 1(Y); Tables 1 and 5 give Z. Table 5's market-5 sequence, Z X Z X X Y Z Z Y Y X Y Z, agrees with Figure 6 in every period except the first. Table 6 has five entries in the Z row and four each in the X and Y rows; Figure 6's label would require five in Y and four in Z. The working paper shows the same split: its Figure 11 (p. 29) gives Y, while its Figures 1, 14, and 16 give Z. This period also informs nobody.

3. Market 3: periods on the 17-period list. Footnote 6 (p. 679) lists market 3's price-test periods as 3, 5, 7, 8, and 10; Table 5's asterisks fall on 3, 5, 6, 8, and 10. Table 3 gives market 3 the same price under both theories in state X (400) and different prices only in state Y (220 against 175), and Figure 4 labels period 6 as Y and period 7 as X, both inside the insider block. Table 4's market-3 row, headed "3 (state Y)", lists efficiencies of 79, 88, 89, 98, and 99 and trading efficiencies of 13, 38, 47, 88, and 94; these match the values printed under Figure 4's panels for periods 3, 5, 6, 8, and 10, whereas period 7 carries 100 and 100. The working paper repeats the footnote (its footnote 3, p. 98), while its Figures 12 and 17 list period 6. Section 3.2 uses period 6; the count of 17 is unchanged.

4. Market 1, periods 6 and 8: disjoint predictions. Table 5's note reads "Allocations predicted by the two models are disjoint," and p. 684 repeats the claim for all 17 asterisked periods. Two of them are in market 1, which Table 3 omits because its insiders received probabilistic information. Tables 1 and 2 imply overlapping sets in both. Types I and II tie for the highest prior value, 283.33 francs. At the clue-conditional probabilities of X for periods 6 and 8 (0.555 and 0.769; Table A4), the highest informed value, 272.2 or 261.5 francs, lies below that tie. PI then leaves the certificates with uninformed members of types I and II, while RE gives them to

all type II traders, so the two sets share the uninformed type II traders. Elsewhere on the list the type with the highest realized dividend never also has the highest prior value, which by Lemma 1(b) is the condition for disjoint predictions. The working paper states the same count on p. 34, while its Figure 17 tabulates only the 15 periods from markets 2–5. The classification of periods is unaffected.

5. Market 4, periods 7 and 8: plotted trades against printed averages. Figure 5 prints average prices of 161 and 164 under these panels, yet every plotted trade lies above 160 and the points cluster near 170. Appendix B of the working paper (p. 92) lists 17 trades in period 7 (160; three at 155; ten at 160; then 165, 170, and 170), which average 160.59, and 12 trades in period 8 (five at 160, four at 165, and three at 170), which average 164.17. These are the printed averages. Measured against the RE line at 175 francs drawn in both panels, each plotted point sits about 10 francs above its Appendix B counterpart, while the step counts agree: period 8 follows a 5–4–3 pattern in both. The working paper's Figure 10 (p. 28) shows the same offset. In the five other market-4 panels checked (periods 3, 4, 5, 10, and 12), the plotted range brackets the printed average. Figure B1 sets the two versions side by side. Both periods are downward periods in the main sample, but the analysis uses the Appendix B trades, which reproduce the printed averages (Table B1), so the offset enters no result. Plott and Sunder's statement that "in all but four periods average price is closer to the RE equilibrium than to the PI equilibrium" (p. 679) holds either way: the two averages lie 14.4 and 10.8 francs from 175, against 49.4 and 45.8 francs from the PI prediction of 210.

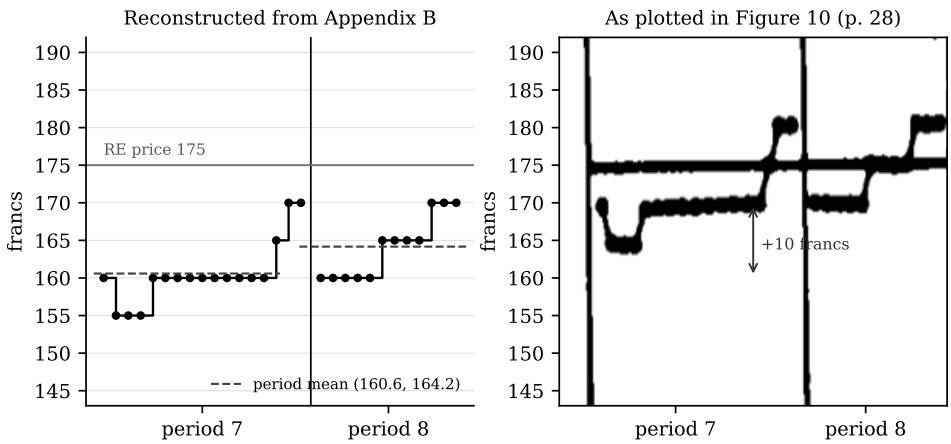


Figure B1: Market 4, periods 7 and 8: listed trades and the published plot. Left: the trades listed in Appendix B of Plott and Sunder (1980), with period means (dashed). Right: the same two panels of working-paper Figure 10, the counterpart of published Figure 5. Both panels share one vertical scale and are aligned on the RE price of 175 francs; the plotted trades sit about 10 francs above the listed trades.

C Robustness of the No-Information Benchmark

This appendix checks the comparison of Section 4. Table C1 lists the 18 no-information periods, and Table C2 repeats the comparison under alternative definitions.

Table C1: The 18 no-information periods

Market	Period	State	Insider block	Trades	First	Mean	Last	Note
1	1	Y	before	17	275	239.4	225	
1	2	Y	before	14	250	263.9	260	
1	3	X	before	14	260	271.8	280	
1	4	Y	before	10	275	279.5	280	
2	1	X	before	11	350	255.0	250	
2	2	X	before	8	250	250.9	260	
2	3	Y	before	14	255	258.2	260	
2	4	Y	before	12	260	264.7	266	
3	1	X	before	12	250	234.2	200	
3	2	Y	before	15	200	217.3	210	

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Market	Period	State	Insider block	Trades	First	Mean	Last	Note
4	1	X	before	8	1000	784.4	200	excluded
4	2	Y	before	14	150	179.6	175	
4	3	Y	before	16	150	153.8	155	
4	4	X	before	16	155	158.1	165	
4	14	Y	after	13	170	158.8	155	
5	1	Z	before	10	150	165.0	170	
5	2	X	before	17	165	172.4	175	
5	3	Z	before	18	170	174.7	175	

C.1 Market 4, period 1

Six of the eight trades in market 4's first period took place at 500 to 1,500 francs (1,000, 1,500, 1,000, 1,000, 900, and 500), and the last two at 175 and 200. Every dividend in market 4 is at most 375 francs, so a certificate bought at 500 francs or more lost money in every state. Plott and Sunder exclude the period from their means (Plott and Sunder, 1982, p. 691) and offer no explanation for the prices. Including it raises market 4's no-information price from 162.6 to 286.9 francs, above μ, which reverses the sign of the difference with downward-period prices; the period mean of 784 francs alone accounts for the change (Table C2).

Table C2: No-information and downward prices under alternative definitions

Market	Version	Periods	$\bar{p}_0$	Downward	Difference	Welch p	Rank-sum p
1	Baseline: period means	1–4	263.7	289.5	+25.9	0.059	0.133
1	Opening prices	1–4	265.0	290.0	+25.0	0.027	0.095
1	Closing prices	1–4	261.2	292.5	+31.2	0.093	0.100
2	Baseline: period means	1–4	257.2	287.2	+30.0	0.372	0.133
2	Opening prices	1–4	278.8	322.5	+43.8	0.163	0.533
2	Closing prices	1–4	259.0	270.0	+11.0	0.829	1.000
3	Baseline: period means	1–2	225.8	171.4	−54.4	0.045	0.095
3	Opening prices	1–2	225.0	179.0	−46.0	0.289	0.160
3	Closing prices	1–2	205.0	170.6	−34.4	0.087	0.079
4	Baseline: period means	2–4, 14	162.6	164.0	+1.5	0.827	0.556
4	Period 1 included	1–4, 14	286.9	164.0	−122.9	0.379	1.000
4	Before the insider block only	2–4	163.8	164.0	+0.2	0.981	0.786
4	After the insider block only	14	158.8	164.0	+5.2	–	–
4	Opening prices	2–4, 14	156.2	158.0	+1.8	0.781	0.796
4	Closing prices	2–4, 14	162.5	168.0	+5.5	0.348	0.368
5	Baseline: period means	1–3	170.7	177.0	+6.3	0.152	0.100

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Market	Version	Periods	$\bar{p}_0$	Downward	Difference	Welch p	Rank-sum p
5	Opening prices	1–3	161.7	188.3	+26.7	0.119	0.100
5	Closing prices	1–3	173.3	175.7	+2.3	0.295	0.157

C.2 No-information periods before and after the insider periods

Every no-information period precedes the insider periods except period 14 of market 4, the last period of that market. Market 4's periods 2–4 average 163.8 francs and period 14 averages 158.8; both are within about 5 francs of the downward-period mean of 164.0. Plott and Sunder describe period 14 the same way: prices "returned to the low levels in period 14 when the no-information condition was imposed again" (Plott and Sunder, 1982, p. 685).

C.3 Trade-level inference

The comparison in Table 5 treats periods as observations. Table C3 uses trades as observations, regressing each trade price on an indicator for a downward period, market by market. With six to nine periods per market, standard errors clustered by period can be far too small, so the table also reports a restricted wild cluster bootstrap with Webb weights (9,999 draws) and a model with a random period intercept. In market 3, all three methods reject equal prices; in market 4, none comes close. Markets 1 and 2 are significant with clustered standard errors and random effects but not with the wild bootstrap, and market 5's difference of about 6 francs is borderline.

Table C3: Trade-level comparison of downward and no-information prices

			OLS			Random period effect	
Market	Trades	Periods	Difference	Cluster p	Wild p	Difference	p
1	77	6	+28.4	0.001	0.139	+26.0	0.048
2	84	6	+32.0	0.041	0.130	+30.2	0.026

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			OLS			Random period effect	
Market	Trades	Periods	Difference	Cluster p	Wild p	Difference	p
3	100	7	−53.4	<0.001	0.024	−54.4	<0.001
4	137	9	+1.8	0.753	0.770	+1.5	0.798
5	94	6	+5.5	0.038	0.039	+6.1	0.061

C.4 Opening and closing prices

Table C2 also compares opening and closing prices instead of period means. The three groups of Section 4.2 remain. In market 3, downward periods open 46 francs and close 34 francs below no-information periods. In market 4, the differences are 2 and 6 francs. In market 5, downward periods open 27 francs above no-information periods but close only 2 francs above them, so their prices fall during the period to the level of no-information closing prices. In markets 1 and 2, downward periods open and close above no-information periods.

C.5 Trader experience

All subjects were students at the Graduate School of Business of the University of Chicago. "In markets 1, 2, and 4 all subjects were inexperienced in that none had previously participated in a laboratory market. Subjects in markets 3 and 5 had participated in one or more of the earlier markets" (Plott and Sunder, 1982, p. 666). Neither the article nor the working paper says which earlier markets these subjects had joined. Experience does not line up with the three groups of Section 4.2: market 3, where prices move from the no-information level to RE, and market 5, where they were already near RE, both had experienced subjects, while market 4, which resembles market 5, had inexperienced subjects. In market 5 the no-information condition was also announced (Plott and Sunder, 1982, p. 670).

D Supplementary Results on Price Paths

D.1 Results by market

Table D1 shows that the pattern of Section 5.2 is not driven by any single market: upward-period prices rise within the period in every market, while downward-period paths are much flatter.

Table D1: Human price positions by market and period type

Market	Period type	Periods	Trades	D_1	$\bar{D}$	D_T	Mean $p_T - p_1$
3	Upward	3	54	0.352	0.802	0.987	+114.3
3	Downward	5	73	0.911	1.081	1.098	-8.4
4	Upward	4	85	0.288	0.577	0.891	+99.5
4	Downward	5	78	1.486	1.313	1.200	+10.0
5	Upward	7	135	0.294	0.659	0.856	+35.7
5	Downward	3	49	0.744	1.092	1.133	-12.7

D.2 Inference and sample

Table D2 reports Welch tests, Mann–Whitney tests, permutation tests within markets, market fixed effects, and alternative samples. The opening-position and franc-change differences survive the within-market permutation and fixed-effect checks in the main sample. With only three market clusters, the clustered fixed-effect result for the path slope is weaker, which is why the main text emphasizes the observed opening and closing values.

Table D2: Human-data inference and sample robustness

Panel	Sample	Measure	Difference	Welch p	MWU p	Within-market p	Clustered FE p
A: inference scheme	markets 3-5, 27 periods	D at first trade	+0.789	0.001	<0.001	<0.001	0.004
A: inference scheme	markets 3-5, 27 periods	D at last trade	+0.251	<0.001	<0.001	<0.001	<0.001
A: inference scheme	markets 3-5, 27 periods	D, period average	+0.507	<0.001	<0.001	<0.001	<0.001
A: inference scheme	markets 3-5, 27 periods	Change across period	-0.538	0.020	0.014	0.014	0.027
A: inference scheme	markets 3-5, 27 periods	Slope on position	-0.385	0.021	0.013	0.019	0.064
A: inference scheme	markets 3-5, 27 periods	Francs moved in period	-73.093	0.001	0.002	<0.001	<0.001
B: sample	Markets 3-5, all 27 periods	D at first trade	+0.789	0.001	<0.001	–	–
B: sample	Markets 3-5, all 27 periods	Francs moved in period	-73.093	0.001	0.002	–	–
B: sample	Flagged period dropped	D at first trade	+0.789	0.001	<0.001	–	–
B: sample	Flagged period dropped	Francs moved in period	-73.093	0.001	0.002	–	–
B: sample	All five markets	D at first trade	+0.196	0.541	0.076	–	–
B: sample	All five markets	Francs moved in period	-61.384	<0.001	0.001	–	–

Notes: Each row compares downward with upward periods; Difference is the downward mean minus the upward mean of the period-level measure. Welch p is from a two-sided two-sample t-test with unequal variances, and MWU p from a two-sided Mann–Whitney U test. Within-market p is from a permutation test that reshuffles the period-type labels among the periods of each market, keeping each market's numbers of downward and upward periods fixed; the two-sided p-value is $(h + 1)/20{,}001$, where h counts the 20,000 reshuffles (seed 20260731) whose absolute difference is at least the observed one. Clustered FE p is for the coefficient on an upward-period indicator in an OLS regression with market fixed effects and standard errors clustered by market; with only three clusters it can overstate significance. In Panel B, the flagged period is market 1's period 2 (Appendix B.1); it lies outside markets 3–5, so dropping it leaves the sample unchanged. p-values below 0.001 are shown as <0.001.

D.3 All five markets

Table D3 adds markets 1 and 2. Pooling all five markets, upward periods open 69 francs below P_{RE} on average and close 15 francs below it. In markets 1 and 2, however, upward-period prices change little within periods (6 francs on average), and downward-period prices open and close above P_{RE}, consistent with Section 4.2.

Table D3: Price positions in markets 3–5, markets 1–2, and all five markets

Markets	Type	Periods	Trades	D_1	$\bar{D}$	D_T	$p_1 - P_{\mathrm{RE}}$	$p_T - P_{\mathrm{RE}}$	$p_T - p_1$
3–5	Downward	13	200	1.09	1.17	1.15	−3.1	−5.4	−2.3
3–5	Upward	14	274	0.30	0.67	0.89	−82.5	−11.7	70.8
1–2	Downward	4	61	−1.27	−0.60	−0.35	52.8	27.8	−25.0
1–2	Upward	5	89	0.44	0.58	0.60	−30.1	−24.1	6.0
1–5	Downward	17	261	0.54	0.76	0.79	10.1	2.4	−7.6
1–5	Upward	19	363	0.34	0.64	0.82	−68.7	−15.0	53.7

D.4 A closeness index

The price position D is signed, so a downward-period price that passes the informed value ($D > 1$) scores above one that stops just short of it. A closeness index that treats deviations on either side alike is

$$A(p) = 1 - \frac{|p - P_{\mathrm{RE}}|}{|P_{\mathrm{RE}} - \mu|} = 1 - |D(p) - 1|,$$

which equals 1 at P_{RE} and falls by the same amount whichever side of P_{RE} the price lies on. In upward periods, where nearly every price lies below P_{RE}, A and D coincide. In downward periods A is lower for two reasons: prices that pass P_{RE} count as deviations, and each franc of deviation counts more because the distance from μ to P_{RE} is short (32.5 to 45 francs, against up to 180 in upward periods). The opening gap between the two kinds of period therefore narrows from 0.79 in D to 0.29 in A, and the closing gap disappears (Table D4). The absolute franc deviations in Table 6 make the same comparison without the second effect: 15 francs at the downward opening against 83 at the upward opening.

Table D4: Price position D and closeness A in markets 3–5

Measure	Downward	Upward	Downward minus upward	Welch p
D, opening price	1.093	0.305	0.789	0.001
A, opening price	0.594	0.305	0.289	0.122
D, period mean price	1.173	0.666	0.507	<0.001
A, period mean price	0.779	0.666	0.113	0.186
D, closing price	1.145	0.894	0.251	<0.001
A, closing price	0.855	0.872	−0.018	0.676

D.5 Distances measured from the no-information price

Section 4 shows that in markets 4 and 5 prices sit well below μ when no one is informed. Table D5 therefore measures the distance to P_{RE} from the no-information price $\bar{p}_0$ as well as from μ, and defines $D^0 = (p - \bar{p}_0)/(P_{\mathrm{RE}} - \bar{p}_0)$. In upward periods, measuring from $\bar{p}_0$ lengthens the distance in markets 4 and 5, and the first trade covers 0.33 to 0.71 of it; the last trade covers 0.83 to 0.99. In the downward periods of markets 4 and 5, $\bar{p}_0$ already lies below P_{RE}. D^0 can still be computed, but reaching P_{RE} from $\bar{p}_0$ would require prices to rise, so it no longer measures downward adjustment, and Table D5 does not report it.

Table D5: Distances to the informed value from the prior value and from the no-information price

Market	Period type	P_{RE}	μ	$\bar{p}_0$	Periods	$P_{\mathrm{RE}} - \mu$	$P_{\mathrm{RE}} - \bar{p}_0$	D^0_1	D^0_T
3	Downward	175	220.0	225.8	5	−45.0	−50.8	0.92	1.09
3	Upward	400	220.0	225.8	3	180.0	174.2	0.33	0.99
4	Downward	175	210.0	162.6	5	−35.0	12.4	–	–
4	Upward	375	210.0	162.6	4	165.0	212.4	0.45	0.92
5	Downward	180	212.5	170.7	3	−32.5	9.3	–	–
5	Upward	245	212.5	170.7	4	32.5	74.3	0.71	0.97
5	Upward	320	212.5	170.7	3	107.5	149.3	0.44	0.83

Notes: – marks entries not reported; see the text.

D.6 Familiar states and the opening window

Table D6 restricts the sample to states that had already appeared in earlier insider periods, which comes closest to Plott and Sunder's qualifier "after some replications" (p. 688). In upward periods the opening position stays below 1 in every sample, while the difference between the two kinds of period shrinks as the sample narrows. The opening window does not depend on using a single first trade either. Averaging over the first tenth of each period's trades gives median deviations from P_{RE} of -5 francs in downward periods and -65 francs in upward periods, and all 14 upward periods remain below P_{RE}.

Table D6: Opening positions in periods with familiar states

Sample	Period type	Periods	D_1	p	D_T
All insider periods	Downward	13	1.093	0.520	1.145
All insider periods	Upward	14	0.305	<0.001	0.894
	Difference	27	0.789	0.001	
State seen in at least one earlier insider period	Downward	10	1.106	0.525	1.123
State seen in at least one earlier insider period	Upward	10	0.579	0.003	0.890
	Difference	20	0.527	0.014	
State seen in at least two earlier insider periods	Downward	7	0.988	0.958	1.114
State seen in at least two earlier insider periods	Upward	6	0.724	0.017	0.933
	Difference	13	0.265	0.277	

Notes: Markets 3–5. "At least one earlier insider period" and "at least two" count earlier insider periods with the same realized state. For a period type, p is from a two-sided t-test of mean $D_1 = 1$; for the difference, p is from a Welch test of equal means.

E Supplementary Material on the LLM Pretest

E.1 Institutions and treatment text

Table E1 sets the original trading institution beside its LLM counterpart. The trading protocol, trade unit, and endowments carry over; timing, quote replacement, the

visible record of past periods, private memory, and belief elicitation do not.

Table E1: The original and LLM market institutions

Feature	Original experiment	LLM markets
Trading protocol	Oral double auction; the latest bid and offer remain outstanding.	Public one-bid, one-offer book with bid, offer, accept, or hold actions.
Timing	Traders can call out quotes at any time during the period.	Random turn order; normally at most three turns per trader per period.
Quote replacement	A later quote can replace the earlier one.	A new bid must improve the standing bid and a new offer the standing offer.
Trade unit	One certificate per accepted bid or offer.	One certificate per accepted bid or offer.
Endowments	Two certificates and 10,000 francs each period; cash and inventory constraints.	Same endowments and constraints, reset each period.
Public record	Bids, offers, and trades written on the blackboard; the record and realized states of the latest four or five periods stayed visible, and market 5 subjects received a photocopy of each period's log.	Public record of every quote and trade in the current period, shown on every turn; no earlier period is displayed.
Private memory	Unrestricted.	Baseline keeps two recent end-of-period notes and three post-trade notes; step 1b and later use one replacing memo of up to 1,200 words.
Belief reports	No elicitation during trading.	Posterior belief, reservation value, and the basis of the belief are elicited with each action.

Notes: Human entries follow Plott and Sunder (1982, p. 666, n. 3; pp. 687–688, 697) and Plott and Sunder (1980, p. 60).

Every LLM agent receives a common instruction with the length of the experiment and the currency, its own dividend schedule, the mechanism for drawing the state and handing out clue cards, statements of common knowledge, the trading rules, and the required action format. The baseline sentence on common knowledge that matters for the disclosure steps reads exactly:

> "No one is told how many investors receive a clue card that is not blank, or which investors they are."

The baseline also says that all nonblank cards in a period carry the same content, that dividends stay the same across periods (except in market 1, where this was not announced), and that a lettered card is always correct. Each turn shows the trader's card, cash, certificates, standing quotes, and the public record. The agent chooses one action; a valid acceptance executes at once, updates holdings, removes the accepted quote, and is announced to all. At the end of a period, dividends and the fixed cost are applied, holdings reset, and a private reflection is stored for later periods.

Step 1 adds the full dividend table and the number of informed traders. For market A, the relevant text is reproduced below, with only the dividend lines compressed into one line; market B uses the same text with its own dividends (Table A2). "Type I" changes to the receiving trader's own type. The complete rendered prompts for steps 0, 1, 1b, 2, and 3 are in `data/llm/prompts/` of the replication package.

> `THE THREE TYPES OF INVESTORS`
> The facts in this section are known to all twelve investors in this market. There are three types of investors, with four investors of each type. Investors of the same type have the same earnings per certificate; investors of different types have different earnings. The three types earn: Type I: X 360, Y 110; Type II: X 330, Y 130; Type III: X 290, Y 160. You are a Type I investor. No one is told which type any OTHER investor is. In a year in which clue cards that are not blank are handed out, exactly two of the four investors of each type receive such a card; the other two receive blanks. No one is told which investors they are, or whether they are the same investors from year to year. There may also be years in which every investor's card is blank, and no one is told which years are which: a blank card looks the same to its holder either way.

Step 1b keeps step 1 and changes three implementation instructions together. The profit objective is:

> "Your objective is to earn as many francs as you can, over the whole experiment rather than in any single year. Judge every decision—what to quote, what to accept, and what to let pass—by what it does to your own francs, and take the one that leaves you with the most."

The added emphasis on clue certainty is:

> "A clue card that carries a letter is always correct. This has no exceptions and no qualifications: if your card carries a letter, the dividend that letter names is the one that WILL be paid at the end of this year. You may treat it as settled."

Finally, the baseline's rolling end-of-period notes are replaced by one persistent private memo. The key instruction is:

> "You keep ONE private memo. No one else will ever read it. It is the only thing you carry forward from year to year, and you will be shown it on every turn you take, so it has to hold everything you want your later self to know: what you have worked out about this market, what the prices have told you and where you read them wrongly, the rules you now trade by, and what you are still unsure of. At the end of a market year, write your memo out again in full. Carry forward everything that is still true, revise what this year has changed, and drop what you now know to be wrong. What you write REPLACES your previous memo completely: anything you leave out is gone, and you will not see it again."

Step 2 adds one of two lines to every trader's card display each period: "This year lettered clue cards have been handed out" or "This year no investor has received a lettered clue card." Step 3 replaces step 2's sentence on fixedness with: "No one is told which investors they are, but they are the same investors in every year in which such cards are handed out." The identities themselves are never disclosed. Thus step 1 to 1b changes three implementation features at once; given the preceding steps, step 1b to 2 varies the announcement of the current information status, and step 2 to 3 varies the disclosure that insiders are always the same traders, not their identities.

E.2 Run inventory and sample selection

Table E2 lists every LLM session. "Traded" reports the number of insider periods with at least one trade over the number scheduled. A period without trade has no first, mean, or last price and drops out of the price summaries, and a paired comparison requires a price in both treatments for the same market, seed, period, and period type. The paired price comparisons are therefore conditional on trade in both treatments. Run IDs use `m7` for market A and `m8` for market B. The background pool includes every session of markets 1–5; the design comparison uses five market-4 sessions and seeds 42–44 of markets A and B. All five disclosure steps use seed 42; steps 2 and 3 and the baseline comparison add seed 45 in A and seed 44 in B. State sequences are fixed within each treatment pair.

Table E2: LLM sessions and sample selection

Run	Mkt.	Step	Seed	Turns	Preset	Realized states	Traded	No-trade	Model
m7_ctrl_42	A	0 / symmetric	42	3	prior redraw	XXXXYXYXYYXXXX	8/9	10	DS-V4-Flash
m7_ctrl_43	A	0 / symmetric	43	3	prior redraw	YXXYXXXXXYYXYY	9/9	–	DS-V4-Flash
m7_ctrl_44	A	0 / symmetric	44	3	prior redraw	XYXYXXXXXXYXYX	9/9	–	DS-V4-Flash
m7_ctrl_45	A	0	45	3	prior redraw	XYYXXXYXYYYYXY	9/9	–	DS-V4-Flash
m8_ctrl_42	B	0 / symmetric	42	3	prior redraw	XXXXYYXXXXXXYY	9/9	–	DS-V4-Flash
m8_ctrl_43	B	0 / symmetric	43	3	prior redraw	XXXXYYXXXXXXYX	9/9	–	DS-V4-Flash
m8_ctrl_44	B	0 / symmetric	44	3	prior redraw	XYYXXYYYXYXYYX	9/9	–	DS-V4-Flash
m7_disc_42	A	1	42	3	prior redraw	XXXXYXYXYYXXXX	9/9	–	DS-V4-Flash
m8_disc_42	B	1	42	3	prior redraw	XXXXYYXXXXXXYY	9/9	–	DS-V4-Flash
m7_lad1b_42	A	1b	42	3	prior redraw	XXXXYXYXYYXXXX	9/9	–	DS-V4-Flash
m8_lad1b_42	B	1b	42	3	prior redraw	XXXXYYXXXXXXYY	9/9	–	DS-V4-Flash
m7_lad2_42	A	2	42	3	prior redraw	XXXXYXYXYYXXXX	9/9	–	DS-V4-Flash
m7_lad2_45	A	2	45	3	prior redraw	XYYXXXYXYYYYXY	9/9	–	DS-V4-Flash
m8_lad2_42	B	2	42	3	prior redraw	XXXXYYXXXXXXYY	8/9	5	DS-V4-Flash
m8_lad2_44	B	2	44	3	prior redraw	XYYXXYYYXYXYYX	9/9	–	DS-V4-Flash
m7_lad3_42	A	3	42	3	prior redraw	XXXXYXYXYYXXXX	9/9	–	DS-V4-Flash
m7_lad3_45	A	3	45	3	prior redraw	XYYXXXYXYYYYXY	8/9	10	DS-V4-Flash
m8_lad3_42	B	3	42	3	prior redraw	XXXXYYXXXXXXYY	9/9	–	DS-V4-Flash
m8_lad3_44	B	3	44	3	prior redraw	XYYXXYYYXYXYYX	9/9	–	DS-V4-Flash
m1_paper_0	1	original	20250725	3	published	YYXYYXYYYXY	4/4	–	DS-V4-Flash
m1_paper_1	1	original	20250726	3	published	YYXYYXYYYXY	4/4	–	DS-V4-Flash
m1_random_0	1	original	20250727	3	prior redraw	YYYXXYYYYYY	4/4	–	DS-V4-Flash
m1_random_1	1	original	20250728	3	prior redraw	XYYYXYYXYYY	4/4	–	DS-V4-Flash
m1_random_2	1	original	20250729	3	prior redraw	YYYYYYYYYXX	4/4	–	DS-V4-Flash
m2_paper_0	2	original	20250735	3	published	XXYYYYXYXYY	5/5	–	DS-V4-Flash
m2_paper_1	2	original	20250736	3	published	XXYYYYXYXYY	5/5	–	DS-V4-Flash
m2_random_0	2	original	20250737	3	prior redraw	XXYYXYYXYYY	5/5	–	DS-V4-Flash
m2_random_1	2	original	20250738	3	prior redraw	YXYYYYXYYYY	5/5	–	DS-V4-Flash
m2_random_2	2	original	20250739	3	prior redraw	YXXYYYXXYYY	5/5	–	DS-V4-Flash
m3_gem_paper	3	original	20250745	3	published	YYYXYYXYXYYX	8/8	–	gemini-3.5-flash
m3_paper_0	3	original	20250745	3	published	YYYXYYXYXYYX	8/8	–	DS-V4-Flash
m3_paper_1	3	original	20250746	3	published	YYYXYYXYXYYX	8/8	–	DS-V4-Flash
m3_random_0	3	original	20250747	3	prior redraw	XYXXXYYYXYYY	8/8	–	DS-V4-Flash
m3_random_1	3	original	20250748	3	prior redraw	XYXYXYYYXXYX	8/8	–	DS-V4-Flash
m3_random_2	3	original	20250749	3	prior redraw	XYYYYXYYYYYY	8/8	–	DS-V4-Flash
m4_paper_0	4	original	20250755	3	published	XYYXYXYYXYXYXY	9/9	–	DS-V4-Flash
m4_paper_1	4	original	20250756	3	published	XYYXYXYYXYXYXY	9/9	–	DS-V4-Flash
m4_random_0	4	original	20250757	3	prior redraw	YXXXXXYYYXYYYY	9/9	–	DS-V4-Flash
m4_random_1	4	original	20250758	3	prior redraw	YYXXXXYYYYYXYX	9/9	–	DS-V4-Flash
m4_random_2	4	original	20250759	3	prior redraw	YYYYXYYXXYXXYY	9/9	–	DS-V4-Flash
m5_paper_0	5	original	20250765	3	published	ZXZXXYZZYYXYZ	10/10	–	DS-V4-Flash
m5_paper_1	5	original	20250766	3	published	ZXZXXYZZYYXYZ	10/10	–	DS-V4-Flash
m5_random_0	5	original	20250767	3	prior redraw	XZYZZXXZZXXXX	10/10	–	DS-V4-Flash
m5_random_1	5	original	20250768	3	prior redraw	ZYYZZYZXYXZZY	10/10	–	DS-V4-Flash
m5_random_2	5	original	20250769	3	prior redraw	YXZXZZXYZZYZX	10/10	–	DS-V4-Flash
m4_r4_paper	4	round check	20250755	4	published	XYYXYXYYXYXYXY	9/9	–	DS-V4-Flash
m4_r5_paper	4	round check	20250755	5	published	XYYXYXYYXYXYXY	9/9	–	DS-V4-Flash
m4_r6_paper	4	round check	20250755	6	published	XYYXYXYYXYXYXY	9/9	–	DS-V4-Flash
m4_r4_random	4	round check	20250757	4	prior redraw	YXXXXXYYYXYYYY	9/9	–	DS-V4-Flash
m4_r5_random	4	round check	20250757	5	prior redraw	YXXXXXYYYXYYYY	9/9	–	DS-V4-Flash
m4_r6_random	4	round check	20250757	6	prior redraw	YXXXXXYYYXYYYY	9/9	–	DS-V4-Flash

E.3 Design comparison by session

Table E3 reports each session behind the comparison of market 4 with markets A and B. Table 7 weights traded periods equally, whereas the directional gap of −0.02 in Section 6.3 first averages each period type within a session and then weights sessions equally. The market-4 and A–B sessions are not matched by seed or state sequence.

Table E3: Mean price position by session in the design comparison

Sample	Run	Upward	Downward	Downward minus upward
Market 4	m4_paper_0	0.229	1.169	+0.940
Market 4	m4_paper_1	0.051	1.704	+1.653
Market 4	m4_random_0	-0.086	1.463	+1.549
Market 4	m4_random_1	0.125	1.019	+0.894
Market 4	m4_random_2	0.178	1.053	+0.875
Markets 1–5	m1_paper_0	0.038	0.122	+0.084
Markets 1–5	m1_paper_1	0.219	1.528	+1.309
Markets 1–5	m1_random_0	-0.868	0.805	+1.673
Markets 1–5	m1_random_1	0.178	0.744	+0.566
Markets 1–5	m1_random_2	-1.665	0.638	+2.303
Markets 1–5	m2_paper_0	0.510	0.141	-0.368
Markets 1–5	m2_paper_1	-0.242	2.872	+3.114
Markets 1–5	m2_random_0	0.092	-0.233	-0.325
Markets 1–5	m2_random_1	0.705	-0.936	-1.641
Markets 1–5	m2_random_2	0.091	2.109	+2.019
Markets 1–5	m3_gem_paper	0.142	1.129	+0.986
Markets 1–5	m3_paper_0	0.183	1.308	+1.125
Markets 1–5	m3_paper_1	0.021	0.894	+0.873
Markets 1–5	m3_random_0	0.150	0.903	+0.752
Markets 1–5	m3_random_1	-0.051	0.557	+0.608
Markets 1–5	m3_random_2	-0.269	1.318	+1.587
Markets 1–5	m4_paper_0	0.229	1.169	+0.940
Markets 1–5	m4_paper_1	0.051	1.704	+1.653
Markets 1–5	m4_random_0	-0.086	1.463	+1.549
Markets 1–5	m4_random_1	0.125	1.019	+0.894
Markets 1–5	m4_random_2	0.178	1.053	+0.875
Markets 1–5	m5_paper_0	-0.399	1.203	+1.602
Markets 1–5	m5_paper_1	-0.640	0.919	+1.558
Markets 1–5	m5_random_0	-0.244	0.870	+1.114
Markets 1–5	m5_random_1	-0.033	0.700	+0.733
Markets 1–5	m5_random_2	-0.213	1.385	+1.598
Markets A and B	m7_ctrl_42	0.421	0.315	-0.105
Markets A and B	m7_ctrl_43	0.404	0.748	+0.343
Markets A and B	m7_ctrl_44	0.304	0.140	-0.164
Markets A and B	m8_ctrl_42	0.525	0.191	-0.335
Markets A and B	m8_ctrl_43	0.511	0.455	-0.056
Markets A and B	m8_ctrl_44	0.252	0.434	+0.181

E.4 Step-by-step disclosure

Each treatment uses the realized state sequence of its seed-matched baseline. Both markets have all five steps at seed 42, and steps 0, 2, and 3 also have a second seed per market. I pair periods by market, seed, period, and period type, average the paired differences within each session pair, and weight session pairs equally (Table E4). A paired price requires trade in both treatments, so treatment-dependent nontrading can change which periods enter a step; Table E7 holds the traded periods fixed across all five steps at seed 42 as a check.

The exact sign-flip test treats session pairs, rather than periods, as exchangeable under the null of no effect, which preserves the shared model history within a session. With two pairs for the early steps and four for the later comparisons, the smallest attainable two-sided p-values are 0.500 and 0.125, so no comparison can reach conventional significance levels. I therefore report magnitudes and the direction of change across pairs.

Table E4: Paired changes in price position across the disclosure steps

Step	Period type	Sessions	Period pairs	Before	After	Change	Positive	Session p
0 to 1: dividends and clue count	Upward	2	11	0.473	0.499	+0.026	1/2	1.000
0 to 1: dividends and clue count	Downward	2	6	0.253	0.260	+0.007	1/2	1.000
1 to 1b: implementation changes	Upward	2	11	0.499	0.580	+0.080	2/2	0.500
1 to 1b: implementation changes	Downward	2	7	0.322	0.555	+0.232	1/2	1.000
1b to 2: period announcement	Upward	2	11	0.580	0.659	+0.080	2/2	0.500
1b to 2: period announcement	Downward	2	6	0.573	0.976	+0.403	1/2	1.000
2 to 3: same insiders	Upward	4	18	0.705	0.823	+0.119	3/4	0.375
2 to 3: same insiders	Downward	4	16	0.997	0.738	-0.259	1/4	0.250
0 to 3: all steps	Upward	4	18	0.309	0.823	+0.514	4/4	0.125
0 to 3: all steps	Downward	4	16	0.470	0.730	+0.261	3/4	0.625

Notes: $D = 0$ at the prior value and $D = 1$ at the informed value. Periods are paired by market, seed, period, and period type, requiring trade in both treatments; paired differences are averaged within session and then equally across session pairs. "Positive" counts pairs with a positive mean change. "Session p" is the two-sided exact sign-flip p-value using session-pair means. Steps involving 1 or 1b use seed 42; steps 2 to 3 and 0 to 3 use two seeds per market. Each row uses its own sample of periods traded in both treatments.

The early steps differ across markets (Table E6). From step 0 to step 1, the mean changes are 0.03 for upward and 0.01 for downward periods, and the downward mean combines −0.31 in A with +0.32 in B. Step 1b adds 0.08 for upward and 0.23 for

downward periods on average. From step 1b to step 2, the downward-period position rises by 0.40 on average, but it rises by 0.88 in A and falls by 0.07 in B, so market A drives the mean; both upward pairs rise, by 0.08 on average. From step 2 to step 3, the upward-period position rises from 0.70 to 0.82, positive in three of four pairs, while the downward-period position falls from 1.00 to 0.74, negative in three pairs. The fall comes from two session pairs, market A at seed 45 (1.02 to 0.24) and market B at seed 42 (0.95 to 0.70); the other two stay near 1. Figure E1 plots the responses of each session pair.

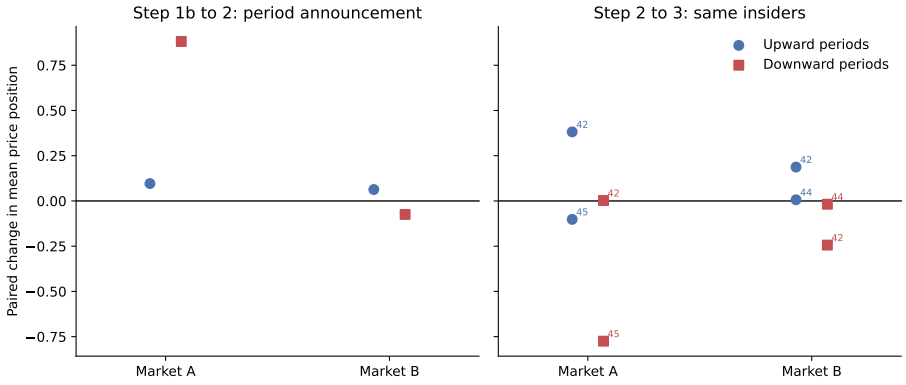


Figure E1: Responses to disclosure by market and session. Each marker is the paired change in the period-mean price position within a market–seed session pair, using periods with trade in both treatments. The left panel compares step 1b with step 2 at seed 42. The right panel compares step 2 with step 3 at two seeds per market, labeled beside the points. The horizontal line marks no change.

Beliefs and allocations change along with several price responses (Table E5). Between steps 1b and 2, uninformed traders' mean probability of the realized state rises from 0.58 to 0.82 in downward periods, transaction efficiency rises from 25.0 to 79.9 percent, and the ratio of insider to uninformed profits falls from 247 to 111 percent. Between steps 2 and 3, the upward-period position in the seed-42 summary rises from 0.67 to 0.94, the true-state probability from 0.86 to 0.91, and transaction efficiency from 76.0 to 97.1 percent. These patterns are consistent with uninformed traders learning the state from prices, but they do not establish the order of cause and effect.

Table E5: Beliefs, allocations, and profits across the disclosure steps at seed 42

Step	Period type	$\bar{D}$	True-state belief	Price basis	TE (%)	Insider profit ratio (%)
0	Upward	0.478	0.743	0.813	55.9	123.9
0	Downward	0.253	0.449	0.673	20.7	555.8
1	Upward	0.513	0.769	0.893	63.5	121.5
1	Downward	0.295	0.469	0.802	-1.3	490.9
1b	Upward	0.588	0.806	0.769	64.9	112.8
1b	Downward	0.493	0.584	0.730	25.0	247.0
2	Upward	0.666	0.860	0.915	76.0	107.4
2	Downward	0.986	0.817	0.868	79.9	110.8
3	Upward	0.942	0.909	0.934	97.1	101.4
3	Downward	0.863	0.834	0.865	76.2	118.8

Notes: $\bar{D}$ is the period-mean price position. True-state belief is the mean probability uninformed traders assign to the realized state. Price basis is the share of their belief reports that cite prices rather than the prior or others' behavior. TE is transaction efficiency, the share of achievable surplus realized relative to the no-trade allocation. The profit ratio divides mean insider profit by mean uninformed profit. Entries weight periods equally across markets A and B at seed 42; effective samples differ by outcome when trade or reports are missing.

Table E6: Disclosure effects by market–seed session pair

Step	Type	Market	Seed	Pairs	Before	After	Change
0 to 1: dividends and clue count	Upward	A	42	5	0.421	0.350	-0.070
0 to 1: dividends and clue count	Upward	B	42	6	0.525	0.648	+0.123
0 to 1: dividends and clue count	Downward	A	42	3	0.315	0.005	-0.310
0 to 1: dividends and clue count	Downward	B	42	3	0.191	0.515	+0.324
1 to 1b: implementation changes	Upward	A	42	5	0.350	0.489	+0.139
1 to 1b: implementation changes	Upward	B	42	6	0.648	0.670	+0.021
1 to 1b: implementation changes	Downward	A	42	4	0.130	0.126	-0.004
1 to 1b: implementation changes	Downward	B	42	3	0.515	0.983	+0.468
1b to 2: period announcement	Upward	A	42	5	0.489	0.586	+0.096

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Step	Type	Market	Seed	Pairs	Before	After	Change
1b to 2: period announcement	Upward	B	42	6	0.670	0.733	+0.063
1b to 2: period announcement	Downward	A	42	4	0.126	1.007	+0.881
1b to 2: period announcement	Downward	B	42	2	1.020	0.945	-0.074
2 to 3: same insiders	Upward	A	42	5	0.586	0.967	+0.382
2 to 3: same insiders	Upward	A	45	4	0.782	0.680	-0.102
2 to 3: same insiders	Upward	B	42	6	0.733	0.920	+0.187
2 to 3: same insiders	Upward	B	44	3	0.718	0.725	+0.007
2 to 3: same insiders	Downward	A	42	4	1.007	1.010	+0.003
2 to 3: same insiders	Downward	A	45	4	1.018	0.243	-0.775
2 to 3: same insiders	Downward	B	42	2	0.945	0.701	-0.244
2 to 3: same insiders	Downward	B	44	6	1.018	1.000	-0.018
0 to 3: all steps	Upward	A	42	5	0.421	0.967	+0.547
0 to 3: all steps	Upward	A	45	4	0.038	0.680	+0.642
0 to 3: all steps	Upward	B	42	6	0.525	0.920	+0.395
0 to 3: all steps	Upward	B	44	3	0.252	0.725	+0.473
0 to 3: all steps	Downward	A	42	3	0.315	1.009	+0.694
0 to 3: all steps	Downward	A	45	4	0.938	0.243	-0.695
0 to 3: all steps	Downward	B	42	3	0.191	0.668	+0.477
0 to 3: all steps	Downward	B	44	6	0.434	1.000	+0.566

Table E7: The five disclosure steps on a common seed-42 period support

Period type	Step	Market–period cells	$\bar{D}$	Change from step 0
Upward	0	11	0.478	+0.000
Upward	1	11	0.513	+0.035
Upward	1b	11	0.588	+0.110
Upward	2	11	0.666	+0.188
Upward	3	11	0.942	+0.464
Downward	0	5	0.302	+0.000
Downward	1	5	0.187	-0.115
Downward	1b	5	0.464	+0.162
Downward	2	5	0.981	+0.680
Downward	3	5	0.886	+0.584

On the common support, the upward-period position is highest at step 3, while the downward-period position peaks at step 2 and falls at step 3, the same pattern as the paired estimates. These entries weight periods equally and differ from the session-weighted estimates in Table E4.

E.5 More turns per trader

Table E8 reports market-4 sessions with four, five, and six turns per trader, run on the published and on one prior-redraw state sequence alongside their three-turn baselines. On the published sequence, the six-turn closing position in upward periods is 0.200, against 0.447 with three turns, and the intermediate horizons do not rise steadily (-0.133 with four turns and 0.176 with five). On the redraw sequence the upward position rises with each added turn but reaches only 0.402 with six. More trading opportunities therefore do not close the upward-period gap in these runs; two state sequences are too few to estimate a stable effect of the horizon.

Table E8: Market 4 with more turns per trader

State sequence	Turns	Period type	Periods	Trades	D_1	$\bar{D}$	D_T
prior redraw	3	Upward	3	58	-0.283	-0.086	0.226
prior redraw	4	Upward	3	64	-0.061	0.034	0.242
prior redraw	5	Upward	3	77	-0.051	0.178	0.303
prior redraw	6	Upward	3	65	-0.061	0.170	0.402
prior redraw	3	Downward	6	142	1.429	1.463	1.152
prior redraw	4	Downward	6	102	-0.038	-0.372	-0.033
prior redraw	5	Downward	6	151	0.929	0.941	1.005
prior redraw	6	Downward	6	162	1.243	1.147	1.205
published	3	Upward	4	74	0.045	0.229	0.447
published	4	Upward	4	88	-0.302	-0.154	-0.133
published	5	Upward	4	92	-0.061	0.126	0.176
published	6	Upward	4	92	0.015	0.117	0.200
published	3	Downward	5	100	1.114	1.169	1.503
published	4	Downward	5	131	1.434	1.392	1.491
published	5	Downward	5	157	1.371	1.520	1.794
published	6	Downward	5	138	0.914	1.120	1.154